

Making animal-friendly and safer diagnostics for human African trypanosomiasis

Evaluation of *Trypanosoma brucei gambiense* variant
surface glycoprotein expression in *Trypanosoma brucei*
brucei

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List of Abbreviations

Ab	antibody
Ag	antigen
AnTaR	Antwerp Trypanozoon antigen Repertoire
AnTat	Antwerp Trypanozoon antigen type
ApoL1	apolipoprotein L1
BSF	bloodstream-form
CATT	card agglutination test for trypanosomiasis
CNS	central nervous system
CSF	cerebrospinal fluid
DBS	dried blood spots
DEAE	diethylaminoethyl cellulose
dpi	days post-infection
DRC	Democratic Republic of the Congo
DSB	DNA double-stranded break
ELISA	enzyme-linked immunosorbent assay
ESAGs	expression site-associated genes
ESs	expression sites
FI	fluorescence intensity
FPLC	fast protein liquid chromatography
gHAT	<i>gambiense</i> human African trypanosomiasis
GPI	glycosylphosphatidylinositol
GPS	guinea pig serum
HAT	human African trypanosomiasis
HMI-9	Hirumi's Modified Iscove's medium-9
HpHbR	haptoglobin-haemoglobin receptor
Hpr	Hp-related protein
iELISA	Inhibition enzyme-linked immunosorbent assay
Ig	immunoglobulin
IMDM	Iscove's Modified Dulbecco's Medium
ITL	immune trypanolysis test
ITM	Institute of Tropical Medicine
kDNA	kinetoplast DNA
LAMP	loop-mediated isothermal amplification
LB	Luria-Bertani
LiTat	Lille Trypanozoon antigen type
mAECT	mini anion-exchange centrifugation technique
MM	matching method
NECT	nifurtimox-eflornithine combination therapy

NHS	normal human serum
NRS	normal rabbit serum
P	passage
PCM	phase-contrast microscopy
PCR	polymerase chain reaction
PI	propidium iodide
Pol	Polymerase
PPV	positive predictive value
PSG	phosphate-buffered saline + 1% glucose
rcf	relative centrifugal force
RDT	rapid diagnostic test
RFU	relative fluorescence units
RIME	repetitive insertion mobile element
RoTat	Rode Trypanozoon antigen type
rpm	revolutions per minute
RT	room temperature
rVSGs	recombinant VSGs
SD	standard deviation
SDS-PAGE	sodium dodecyl sulphate polyacrylamide gel electrophoresis
SE	single-expressors
SL	spliced leader
spp.	species
SRA	serum resistance-associated gene
<i>T. b.</i>	<i>Trypanosoma brucei</i>
TBR	<i>Trypanosoma brucei</i> repeat
TgsGP	<i>T. b. gambiense</i> -specific glycoprotein gene
TLF	trypanolytic factors
TPP	Target Product Profile
VAT	variable antigen type
VSG	variant surface glycoprotein
WBC	white blood cells
WHO	World Health Organization

Abstract

African trypanosomes are extracellular protozoan parasites of the genus *Trypanosoma* which are transmitted by bloodsucking tsetse flies. Human African trypanosomiasis (HAT), also called sleeping sickness, is mainly caused by *Trypanosoma brucei gambiense* in West and Central Africa. After infection, the parasites multiply and spread through the bloodstream and lymphatics, eventually crossing the blood-brain barrier, thereby causing severe neurological symptoms [1].

In order to maintain a chronic infection in the mammalian host, trypanosomes must evade the immune system. In a process called antigenic variation, the parasites can switch the composition of their variant surface glycoprotein (VSG) coat. Switching the expressed surface antigen ensures that a subpopulation of the parasites avoids being recognised and killed by the host's immune system, thus establishing a new parasitic wave in the blood [2].

Diagnosis of HAT mainly relies on serological testing of antibodies against these VSGs using point-of-care tests such as CATT and RDTs. These tests are often based on the *T. b. gambiense* VSGs LiTat 1.3 and LiTat 1.5, which the patient's antibodies can often recognise early in infection. However, over the last few years, substantial advances in disease control and treatments have caused a significant decline in the number of HAT cases, making it a realistic goal to interrupt disease transmission by 2030 [3]. As a result of the decrease in prevalence, diagnostic tests should be highly specific to avoid generating an overwhelming number of false positives [4]. The most specific serological test is the immune trypanolysis test (ITL). Here, live *T. b. gambiense* clones expressing the VSGs LiTat 1.3 or LiTat 1.5 are exposed to human serum, resulting in opsonisation and antibody-mediated complement lysis if specific antibodies are present in the serum [5, 6]. However, the ITL has some considerable downsides. Highly virulent live *gambiense* trypanosomes need to be maintained in rodents, therefore consuming laboratory animals and putting laboratory personnel at a biohazard risk. In addition, the readout occurs by microscopy, which limits the number of samples that can be read out per time unit and both variants need to be tested to identify cases with high sensitivity. Moreover, the test is only performed at the three WHO Collaborating Centres [7].

This project aimed to address these drawbacks using transgenic *Trypanosoma brucei brucei* strains expressing *gambiense*-VSGs. This non-human infective species can be relatively easily cultivated *in vitro* and genetically modified, making it suitable to integrate fluorescent reporter genes into its genome. This would allow a more rapid multiplex detection using the Spark CYTO 400 cell imager.

Here we show that (1) the transgenic *T. b. brucei* strains react similarly to *T. b. gambiense* strains in panels of rabbit sera in conventional ITL, (2) we can generate green (TagGFP2) and red (DsRed) fluorescent variants, without altering this phenotype, (3) we can automate the readout by combining both green and red variants in a Spark CYTO 400 assay and finally, (4) the automated results are concordant to the ITL when tested on human sera.

The successful validation and *in vitro* cultivation of the transgenic *T. b. brucei* strains pave the way for safer and more animal-friendly serodiagnosis. Moreover, the automated high-throughput ITL using fluorescent *T. b. brucei* LiTat 1.3 and LiTat 1.5 trypanosomes showed promising results that could potentially enhance HAT diagnostics and contribute to disease elimination efforts.

1. General introduction

1.1 Taxonomy

The order Kinetoplastida consists of an extensive number of protozoan parasites. Several belong to the family Trypanosomatidae (Supporting figure 1), which are parasites of invertebrates, vertebrates, and plants. *Leishmania* and *Trypanosoma*, two genera of these trypanosomatids, comprise numerous species known to be causative agents for some detrimental neglected tropical diseases in humans [8-11]. Examples are Leishmaniasis, Chagas' disease and African sleeping sickness, caused by *Leishmania* species, *Trypanosoma cruzi*, and *Trypanosoma brucei* sp., respectively. *Leishmania* and *Trypanosoma* are dioxenous species using hematophagous insects as their vector [10]. The genus *Trypanosoma* is divided into two groups: stercoraria and salivaria, based on how the parasite is transmitted by the vector. *T. brucei* consists of three subspecies: *T. b. brucei*, *T. b. rhodesiense* and *T. b. gambiense*. Together with the non-tsetse transmitted *T. evansi* and *T. equiperdum*, these species are collectively classified in the subgenus Trypanozoon [12].

1.2 Epidemiology

In sub-Saharan Africa, *T. b. gambiense* and *T. b. rhodesiense* are responsible for sleeping sickness, also called human African trypanosomiasis. Hence, trypanosomes rely on the blood-sucking tsetse fly vector (*Glossina*) for their transmission; the distribution of the tsetse fly highly determines the parasite spread, usually limited to known HAT-foci. HAT occurs in an acute and a chronic form, caused by *T. b. rhodesiense* and *T. b. gambiense*, respectively [1, 13]. Chronically infected patients mainly occur in West and Central Africa, whereas acute infections are limited to Eastern and Southern Africa. As a result of different interventions over the past few years, the number of HAT cases has dropped dramatically [4, 14]. The World Health Organization (WHO) reports that, between 1999 and 2021, the number of newly reported *T. b. gambiense* cases significantly dropped by 97%, from 27 862 to 747. Similarly, the new acute cases fell by 91%, from 619 to 55. According to the WHO, the chronic form represented more than 93% of the cases in 2021. Despite the efforts, both *T. b. gambiense* and *T. b. rhodesiense* remain endemic in certain countries in sub-Saharan Africa (Figure 1). The majority of *gambiense* HAT (gHAT) cases remain in the Democratic Republic of the Congo (DRC) [1, 14]. Besides being pathogenic to humans, trypanosomiasis also occurs in domestic animals, called 'nagana'. Here, the three foremost species are *T. congolense*, *T. vivax* and *T. b. brucei*. In addition to *T. vivax*, two other trypanosome species are found outside Africa, i.e., *T. evansi* (surra) and *T. equiperdum* (dourine). They all lead to severe animal mortalities, affecting livestock and having a profound economic impact in sub-Saharan Africa [13, 15, 16].

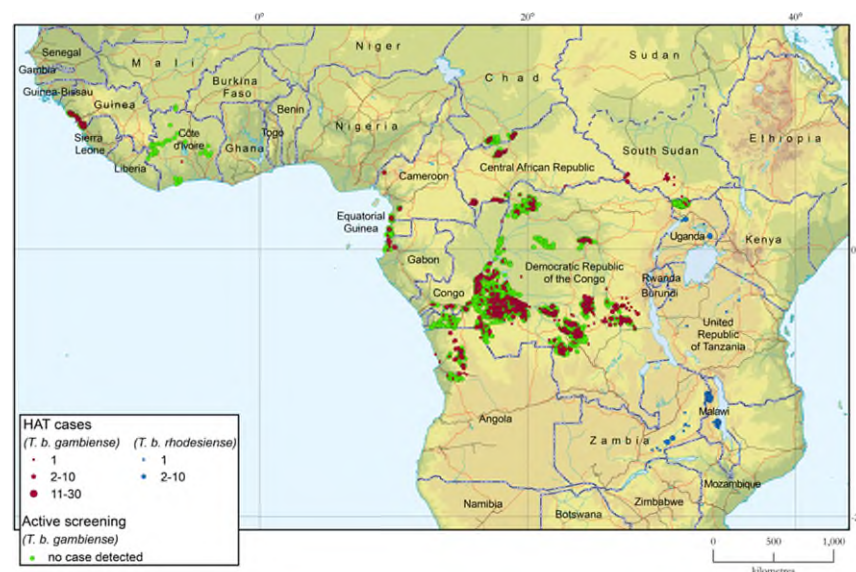


Figure 1: Geographical distribution of HAT cases (2019-2020) [17].

the parasite population resides in the skin and adipose tissue. These sites possibly provide a reservoir for sustaining transmission as they may be immune-privileged [2]. Some parasites will transform into a stumpy form in the bloodstream, thereby being arrested in the G1/G0 phase. Slender trypomastigotes ensure chronicity by continuing replication via binary fission. Utilising quorum sensing, slender forms will eventually differentiate into stumpy forms. These can then be ingested by tsetse flies, transform into procyclic trypomastigotes in the fly's midgut and multiply by binary fission. Parasites then migrate to the proventriculus of the vector via the ectoperitrophic space and differentiate into epimastigotes. In the salivary glands, the epimastigotes attach to the epithelium and multiply by binary fission, generating cells that differentiate into metacyclic trypomastigotes. This is the final form in the tsetse fly and is preadapted to the mammalian host [1, 19, 27].

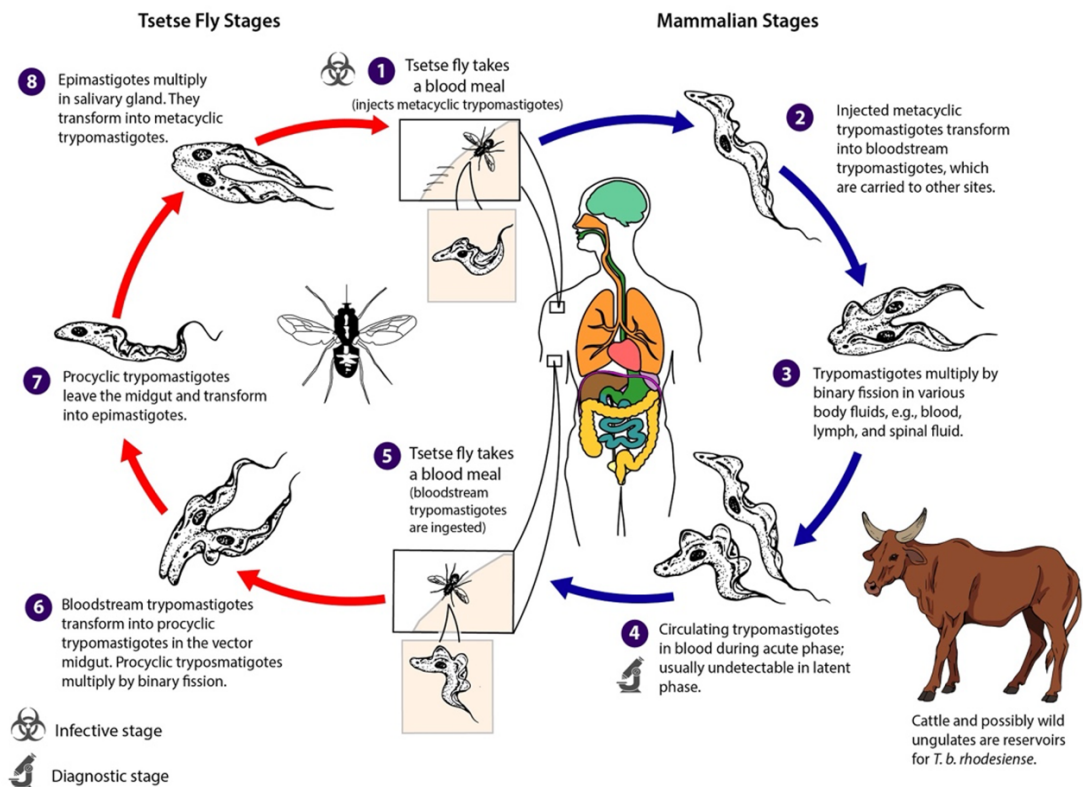


Figure 3: Life cycle of African trypanosomiasis [27].

1.3.3 The tsetse fly vector

By being responsible for transmitting both human and animal African trypanosomiasis, tsetse flies can cause substantial morbidity and mortality in humans and hinder sustainable livestock development. Tsetse flies belong to the genus *Glossina*, which can be divided into three groups depending on their habitat [28, 29]. *G. palpalis* species are found on foliage near rivers and lakes, *morsitans* group species occur in savannah woodlands, and the *fuscus* group prefers forested mangroves [29]. All tsetse fly species can transmit trypanosomes. However, gHAT is mainly linked to tsetse flies of the *palpalis* group, whereby transmission of *T. b. gambiense* happens from human to human (anthroponosis). In contrast, *rhodesiense* HAT is transmitted by *morsitans* spp. and can rely on domestic and wild animals as reservoirs (zoonosis) [28].

If more than one strain is present in the tsetse fly, genetic exchange between trypanosomes can occur, leading to more genetic diversity [12]. The duration of the cycle in the fly takes approximately four weeks. About 0.01% of the tsetse flies are infected in the salivary glands and thus able to transmit *T. b. gambiense* metacyclic trypomastigotes. However, a tsetse fly with a lifespan of three months, feeding every three days, is able to infect several people during that period [1, 19, 27].

1.4 Clinical features of HAT

Generally, the clinical presentation of African sleeping sickness occurs in two distinct stages. A haemolymphatic stage is followed by a meningoencephalitic stage [30]. Clinical manifestations also depend on the *T. brucei* subspecies, as there exist differences in pathogenicity and virulence. *T. b. gambiense* induces a chronic illness of approximately three years, whereas *T. b. rhodesiense* is rapid, with death usually within six months. Nevertheless, both subspecies are fatal if untreated [1, 29]. The infection starts with an infected tsetse bite, possibly causing an inoculation chancre within 2-3 days which is more frequently seen in *T. b. rhodesiense* cases [1, 31].

The first stage typically starts after 1-3 weeks. Intermittent fever episodes accompany headaches, pruritus, weight loss and arthralgia. These symptoms usually disappear after a few days and reappear weeks later. With the progression of the disease, trypanosomes spread to the lymphatics and symptoms such as lymphadenopathy, hepatosplenomegaly, oedema, and infertility can occur. Posterior cervical lymphadenopathy is a typical sign of gHAT known as Winterbottom's sign. Symptoms in this stage tend to be non-specific, which means that a considerable amount of time can pass before appropriate treatment is started [1, 30, 31].

After months or years, the second stage arises because the trypanosomes cross the blood-brain barrier and invade the central nervous system (CNS). Early-stage symptoms now come along with various symptoms related to the CNS. The most characteristic symptom is the disturbance of the circadian rhythm, hence 'sleeping sickness', causing daytime somnolence and nocturnal insomnia. Additionally, motor symptoms, such as gait disturbance, tremor, motor weakness, and speech disorders can occur. Psychiatric symptoms are also common with hallucinations, anxiety, apathy, stereotypic behaviour, manic episodes and dementia [1, 30, 31]. If untreated, the disease can proceed to stupor and coma. Compared to gHAT, the clinical representation in *T. b. rhodesiense* cases is much more acute, with cardiac symptoms such as myocarditis that are more severe [32].

Travellers from non-endemic countries have a shorter incubation period and clinical manifestations that are more acute. Chancres and gastrointestinal symptoms such as jaundice occur more frequently. When untreated, the disease can rapidly affect multiple organs and cause disseminated intravascular coagulation [1, 32].

1.5 Treatment

Previously the treatment of HAT was subspecies and stage-specific. Due to the toxicity of certain drugs, it was crucial to distinguish between early- and late-stage HAT. This was done by counting the number of white blood cells (WBC) in the CSF derived from a lumbar puncture [30]. Pentamidine was the first-choice drug for the treatment of early-stage gHAT given its high cure rate (93-98%) and lower toxicity than suramin [33, 34]. Suramin can also be used for the haemolymphatic stage, but nowadays, it is primarily used for *T. b. rhodesiense* diseases [1]. For drugs to be effective in late-stage treatment, they must be able to cross the blood-brain barrier [33]. For years, the first-line treatment for this stage of gHAT was the nifurtimox-eflornithine combination therapy (NECT). Melarsoprol is an arsenical derivative with a painful administration and a high fatality rate of around 6%. Due to these severe adverse effects, melarsoprol is only used in treating late-stage *T. b. rhodesiense* [1, 34].

However, after completing the phase III clinical trial in 2020, fexinidazole was found to be effective for both stages of gHAT. Ascribed to its oral administration, the treatment is less labour-intensive and more cost-effective. Additionally, it shows equal efficacy compared to older first-line drugs when taken correctly during its 10-day treatment schedule. Therefore, the older drugs are now being replaced by fexinidazole [33, 34]. More recently, in November 2022, a single-dose oral drug named acoziborole was announced to be effective in treating sleeping sickness. With a favourable safety profile and a success rate of 95% in late-stage and 100% in early-stage patients, acoziborole opens the possibility of a simple and effective 'screen-and-treat' approach (this means no parasitological confirmation required) [35].

1.6 Diagnosis of *gambiense* HAT

1.6.1 Parasitological diagnosis

Clinical signs are often not specific enough and thus mistaken for other diseases. Therefore, diagnosis is an important factor in finding positive cases. Parasitological diagnosis of trypanosomes in blood, lymph, or cerebrospinal fluid (CSF) using phase-contrast microscopy (PCM) provides direct evidence of infection. Considering the usually higher level of parasitaemia in *T. b. rhodesiense* patients, direct visualisation is often feasible. However, this is not the case for *gHAT*, where parasite loads are generally lower, and thus direct visualisation might be moderately sensitive [3, 5, 7]. To concentrate the trypanosomes that are present in the blood, mini anion-exchange centrifugation technique (mAECT) is applied. The blood is passed over a diethylaminoethyl cellulose (DEAE) column, separating the trypanosomes. These are then concentrated by low speed microcentrifugation in a transparent collector tube and examined by microscopy. With this concentration method, detection of less than 50 trypanosomes/mL is achievable [36].

1.6.2 Molecular detection

Loop-mediated isothermal amplification (LAMP) and polymerase chain reaction (PCR) rely on the enzymatic amplification of specific DNA or RNA sequences. Multi-copy targets are the repetitive insertion mobile element (RIME), *T. brucei* repeat (TBR), 18S ribosomal RNA gene and the spliced leader (SL) sequence. Identification of *T. b. gambiense* relies on the *T. b. gambiense*-specific glycoprotein gene (TgsGP), while the marker for *T. b. rhodesiense* is based on its serum resistance-associated gene (SRA). However, these methods are not often used for active screening due to their high cost, complexity and generally low sensitivity [4, 5].

1.6.3 Serological diagnosis

1.6.3.1 Antigenic variation and immune evasion

As a result of their extracellular life, trypanosomes are constantly attacked by the host's immune system [4]. To secure their transmission, bloodstream trypanosomes need mechanisms to evade this hostile environment. In the case of *T. brucei*, this is done by means of antigenic variation [37]. BSF trypanosomes are covered with a coat consisting of 10^7 VSG copies [12, 24]. VSGs are homodimers of two 50-60 kDa subunits attached to the plasma membrane by a glycosylphosphatidylinositol (GPI) anchor [38]. This coat ensures that invariant proteins are protected against recognition by the host's immune system and prevents complement-mediated lysis [24]. Although trypanosomes have a genomic repertoire of >1000 VSG-encoding genes, they only express one variable antigen type (VAT) at a time unless the parasite is in the process of switching its VSG [23, 39]. Antigenic variation involves switching the expression of the VSG, avoiding recognition and killing of a subpopulation by the immune system [2]. During the increase of parasitaemia, most trypanosomes express the same VSG, called a homotype (Figure 4). Subsequently, antibodies will be made against this VAT, decreasing the parasitaemia. However, some parasites will express distinct VSGs and thus be able to multiply. As a result, one homotype will overgrow the others, leading to a new wave [12]. The VSG coat also allows the removal of VSG-specific antibodies bound to the parasite's surface. As stated before, all endo- and exocytosis happens at the flagellar pocket. Therefore, when antibodies are bound to the VSG, they will be degraded during endocytosis of the Ab-VSG complex, and the cleared VSGs will be shuttled back to the surface [23].

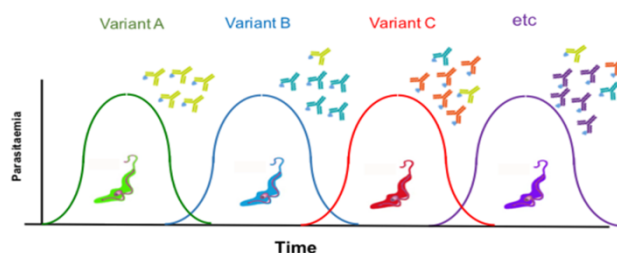


Figure 4: Antigenic variation in *Trypanosoma brucei*.

Progressive waves of parasitaemia during an infection. Each wave consists of a population expressing a single VSG coat, leading to specific antibodies against that variant. However, in real infections, each wave is composed of trypanosomes with different VSGs [2].

Serodiagnostic methods build on the principle of antigen recognition by the antibodies directed against them in the serum of infected patients. Therefore, for VATs to be efficient in diagnostic tests, their surface epitopes should be shared with the ones appearing at an early stage of infection. Furthermore, the used VATs should have a wide range of distribution to overcome the diversity in antigen repertoires. Van Meirvenne *et al.* showed that some VATs, such as Lille Trypanozoon antigen types 1.3 and 1.5 (LiTat 1.3 and LiTat 1.5) fulfil these criteria: they are predominant in *T. b. gambiense*, meaning they are recognised by almost all gHAT patients early in infection and are broadly distributed [6, 39].

Serological tests are preferred to screen for *T. b. gambiense* infection as these tests are not linked to the patient's parasitaemia. A common downside of serological tests is that they are ineffective in differentiating between current or past infections [3, 5, 7].

1.6.3.2 Card agglutination test for trypanosomiasis (CATT)

The card agglutination test for trypanosomiasis (CATT), produced by the Institute of Tropical Medicine (ITM), has become the standard technique for large-scale screening due to its simplicity, reliability, and low cost. The test uses blood, plasma, or serum to detect the presence of anti-trypanosome antibodies. The antigens (Ag) used are the freeze-dried *T. b. gambiense* LiTat 1.3 and LiTat 1.5 parasites fixed with formol and stained with Coomassie blue. On a test card, one drop of blood (or serum/plasma) is added to a drop of the CATT-Ag. The mixture is spread on the test zone and rocked on a flat rotator. After a few minutes, a visible agglutination reaction indicates the presence of anti-parasite antibodies [4, 5]. One test card permits the testing of up to ten samples simultaneously, allowing to screen hundreds of individuals in one day, even on the field [40]. Since its introduction, the CATT has been widely used. However, now that the prevalence of gHAT cases has significantly dropped, CATT suffers from a low positive predictive value (PPV). With a specificity of 98% and a prevalence of 0.01%, the observed PPV is an unsatisfactory 5% [1, 4]. In addition, in some countries like DRC, a positive CATT still has to be followed by microscopy to confirm the infection [4, 7].

1.6.3.3 Rapid diagnostic tests (RDTs)

Two companies develop rapid diagnostic tests (RDTs) for *T. b. gambiense*. Both the HAT Sero-K-SeT (CORIS BioConcept, Gembloux, Belgium) and the SD Bioline HAT 1.0 (Standard Diagnostics, Yongin, South Korea) make use of natively purified LiTat 1.3 and LiTat 1.5 VSGs. A drop of blood and a buffer solution is placed on a cassette with a test strip. The appearance of a control line and one or two test lines is considered a positive test [4, 5]. The most significant advantage over CATT is that no electricity or laboratory infrastructure is needed [1]. Unfortunately, RDTs have the same, or even worse, problems regarding their PPV and discrimination between active or cured cases [4].

1.6.3.4 (Inhibition) ELISA

During the post-elimination phase of HAT, countries shift from active screening to monitoring tools to assess sustained absence or re-emergence of the disease. In this post-elimination stage, enzyme-linked immunosorbent assays (ELISA) are expected to gain more importance due to their high sensitivity and specificity. In these tests, plates are coated with *T. b. gambiense* antigens, blocked with a protein solution to avoid non-specific binding, and then the samples are added. After incubation, plates are washed and incubated with anti-human IgG. After washing, substrate and a chromogen solution are added. If antibodies against *T. b. gambiense* are present, anti-human IgG will bind to them and cause a colour reaction [5, 41, 42]. Furthermore, inhibition ELISA (iELISA) can also be used. Here, the binding of monoclonal antibodies to specific epitopes of *T. b. gambiense* will be inhibited by antibodies of the gHAT patients directed to the same epitopes [7]. ELISA allows high-throughput detection, testing at least three times more samples than the immune trypanolysis test. The biggest drawback is that ELISA still needs laboratory equipment and proper storage and is more time-consuming than CATT or RDTs [5, 7].

1.6.3.5 Immune trypanolysis test (ITL)

The WHO recognised the immune trypanolysis test (ITL) as a reference test for *T. b. gambiense* due to its high specificity and sensitivity. ITL is used to confirm suspected cases diagnosed by other serological techniques when microscopy or molecular tests are ineffective [5, 7]. Live *T. b. gambiense* clones expressing LiTat 1.3 or LiTat 1.5 VSGs are exposed to human serum. The attachment of immunoglobulin (Ig) M or IgG to the parasite's surface allows the binding of complement components to the Fc-region, resulting in the opsonisation of the trypanosomes and, finally, antibody-mediated complement lysis [43, 44]. Therefore, guinea pig serum (GPS) is added to the test due to its high complement activity [6]. The readout of the test involves the observation of lysis using phase-contrast microscopy (PCM). The test is positive when more than 50% of the parasites are lysed. Each sample is subjected to testing against the two LiTat variants, and when a sample is determined to be positive on one variant, it confirms the serological status of the patient.

Like ELISA, ITL can be performed on serum, plasma, and dried blood spots (DBS). The use of DBS simplifies the transportation and storage of patient samples to a great extent. In contrast to serum or plasma, DBS can be shipped at ambient temperature, which is much more feasible in rural settings. However, fewer antibodies can be retrieved from the filter papers, lowering the number of positive results [45].

The ITL has certain limitations: the test is only performed at the three WHO Collaborating Centres on HAT (Institute of Tropical Medicine Antwerp, Belgium, Institut National de Recherche Biomédicale, DRC, and Centre International de Recherche-Développement sur l'Élevage en zone Subhumide, Burkina Faso), has a low throughput, high cost, and demands the *in vivo* propagation of *T. b. gambiense*. Especially the latter is a significant disadvantage of this method, as it means a serious biohazard for the laboratory personnel and uses laboratory rodents. Moreover, specific training and quality assurance is needed to prevent errors due to insufficient blood on the filter paper or inconsistency with the standard protocol [5-7, 46, 47].

1.6.4 The use of recombinant VSGs

The biohazard of cultivating virulent live trypanosomes in laboratory animals results in high demand for recombinant VSGs (rVSGs). Several systems have already been considered to recombinantly express these antigens and replace the native VSGs [48].

Prokaryotic cells such as *Escherichia coli* are often preferred for expressing foreign proteins. This is because they offer rapid growth and high-level expression and require inexpensive growth media. However, they lack the post-translational machinery needed for modifications like protein glycosylation [49, 50]. Compared with bacteria, the use of yeasts like *Pichia pastoris* has the benefit that post-translational modification is present. Thus, allowing the proper folding of the VSG, which is vital for its diagnostic value. Furthermore, this system allows the production of high yields of recombinant proteins and glycosylation, similar to the process in mammalian cells. Therefore, *P. pastoris* has been found to express soluble VSGs effectively [51]. Alternatively, *Spodoptera frugiperda* cell lines, derived from the ovaries of the fall armyworm, are often used to produce recombinant proteins [52]. Currently, these cells are used for the production of recombinant LiTat 1.3 and LiTat 1.5 (personal communication with Philippe Büscher). However, transfection of genetic material in these cells is not integrated into the genome, causing a transient expression. Therefore, the production of proteins is time-limited and lost during cell division [53, 54]. As recombinant expression aims to produce material that is as close to the original VSG as possible, the related trypanosomatid *Leishmania tarentolae* has already been used. *L. tarentolae* allows for simple, cheap and efficient production of trypanosome proteins that are as effective as the native antigens [48]. To date, the homologous expression of rVSGs in *T. b. brucei* has not been tested thus far.

1.7 Variant surface glycoprotein (VSG)

1.7.1 VSG expression

The genome of trypanosomes contains more than 1000 VSG genes and pseudogenes [55]. The vast majority of these genes are present in transcriptionally silent subtelomeric regions. The expression of VSG in a monoallelic manner is done by only transcribing the VSG gene present in one of 15 telomeric VSG expression sites (ESs). The ESs are polycistronic, with expression site-associated genes (ESAGs) upstream of the VSG. Interestingly, ESs are transcribed by RNA Polymerase (Pol) I (which only transcribes ribosomal DNA in other eukaryotes) instead of RNA Pol II [2, 12, 23]. The ability of *T. brucei* to use Pol I to transcribe protein-coding genes is possible because a capped Pol II-derived spliced leader RNA is added to the 5' ends of its mRNAs through trans-splicing [56, 57]. The telomeric location of the ESs allows for the easy exchange of the VSG genes and the generation of genetic diversity [58].

1.7.2 Rate and mechanisms of VSG switching

The rate at which African trypanosomes switch their VSGs varies. Some studies indicate a rate of 10^{-3} switches per population doubling for *in vivo* infections, while *in vitro* studies suggest a lower rate, ranging between 10^{-6} and 10^{-5} . The origin behind this variation is presumably a result of different parasite strains, although environmental factors could also influence the switching rate. For instance, the rate of VSG switching upon transmission through a tsetse fly is found to increase. Hence, the switching rate of natural infections remains unclear as most findings are based on rodent models or *in vitro* cultures [23, 59].

To change the expressed VSG, trypanosomes rely on several switching pathways (Figure 5). The first mechanism is turning off the transcription of the active ES while a silent one is activated/expressed. This process is called *in situ* switching and does not involve DNA rearrangements [60]. Furthermore, VSG switching can also occur via recombination, in which two types can be distinguished. In crossover events, both the silent and active VSG genes exchange place. On the other hand, gene conversion requires the duplication of a silent VSG into the active ES, thereby replacing the active VSG. Fragments of different silent VSG genes have also been shown to recombine in the active ES to form a new mosaic VSG gene [55]. Finally, the loss of the entire active ES can also occur with an *in-situ* switch, a pathway called ES loss + *in situ* [61]. The exact process underlying the initiation and regulation of VSG switching is still unclear. DNA double-strand breaks (DSBs) within the homologous sequences next to the active VSG are thought to induce VSG switching. Previous studies have shown that artificially induced DSBs upstream of the active VSG led to a 250 times increased frequency of VSG switching. However, further research is still needed to unravel these mechanisms [55, 62].

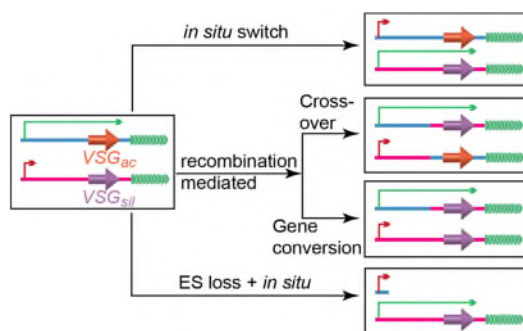


Figure 5: VSG switching pathways.

The active VSG (VSG_{ac}) can be switched into a silent VSG (VSG_{sil}). Green and red arrows represent the active and silent ES promotor, respectively [55].

1.7.3 The expression of ectopic VSGs

Effective antigenic variation in *T. brucei* relies on the monoallelic expression of a single surface antigen and the continuous switching of their VSG during chronic infection. To achieve this, trypanosomes must generate high levels of VSG expression from a single gene. Ridewood *et al.* examined the genomic features that enable such high levels of expression and assessed the ability to express ectopic VSGs located in different genomic positions.

The study found that functional levels of VSG could be expressed after an ectopic VSG 117 construct was inserted immediately downstream of the active ES in the *T. b. brucei* Lister427 single-marker (SM221) cell line. The generated double-expressor line (SM221/117) expressed the ectopic VSG 117 alongside the endogenous VSG 221. The expression levels of VSG 221 were found to be reduced to approximately 50% of wild-type levels, and an inverse correlation was observed between the expression levels of VSG 117 and VSG221. The telomeric VSG 221 was then replaced with a blasticidin resistance gene, resulting in a single-expressor (SE) that exclusively expressed VSG 117 at increased levels. Other genomic locations were also tested, including a Pol II-transcribed α - β -tubulin locus. However, the expression of high levels of ectopic VSG was only achievable when Pol I transcription units such as the ES and the ribosomal DNA were used. Additionally, high levels of expression were exclusively obtained when the VSG was flanked with a VSG 3'UTR and a conserved 16-mer sequence was confirmed to be essential for stabilising the VSG transcript [63].

Using this knowledge, dr. Stijn Rogé (ITM) and MSc Simaa Rene Vigha created two strains of *T. b. brucei* expressing the *gambiense*-specific VSGs LiTat 1.3 and 1.5. Starting from the *T. b. brucei* Lister427 VSG 221 strain, the blasticidin-resistance gene was replaced by a puromycin-resistance gene and a LiTat variant, making double-expressor strains with an active VSG 221 ES unable to switch their VSGs in the presence of puromycin. Then, to acquire LiTat 1.3 or 1.5 SEs, the VSG 221 was knocked out, and a blasticidin-resistance gene was added to select the SEs (for details about the constructs, see Figure 6).

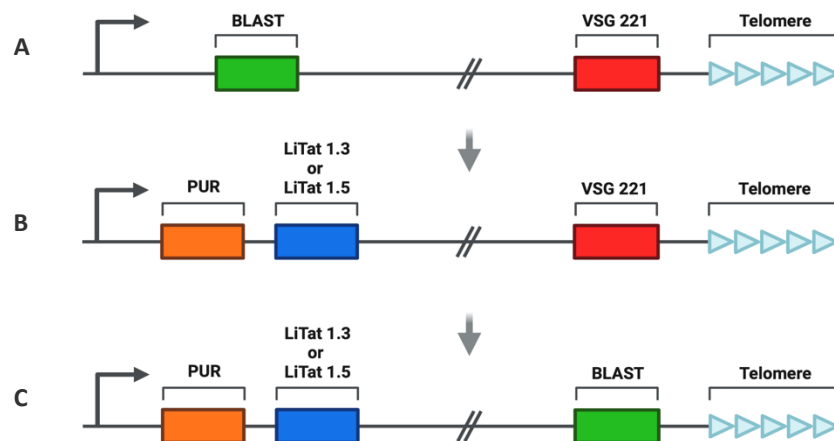


Figure 6: Schematic representation of the ESs in *T. b. brucei* Lister427.

(A) The active VSG 221 expression site with a blasticidin resistant gene (BLAST) downstream of the promoter. (B) Double expressor with an active VSG 221 ES and LiTat 1.3 or 1.5, including puromycin (PUR) resistance gene downstream of the promoter. (C) LiTat 1.3 or 1.5 single expressor after knockout of VSG 221, containing both puromycin and blasticidin-resistance genes.

2. Research question and objectives

Despite the achieved decrease in HAT incidence in sub-Saharan Africa, diagnosis remains vital in the process towards the elimination of transmission. However, the production of antigens used in serodiagnosis relies on cultivating highly virulent parasites in rodents; in addition to the high demand for laboratory animals, this also implies a substantial biohazard safety risk. In the case of the ITL, this is added to the fact that the test has a low throughput, as the parasite lysis must be assessed by microscopy [5, 7]. Here, the project aims to address these drawbacks of serological tests, with particular emphasis on the ITL. Therefore, we will evaluate the diagnostic potential of fluorescent *T. b. brucei* strains expressing LiTat 1.3 or LiTat 1.5.

Research question: Can *in vitro* cultivated fluorescent *T. b. brucei* strains expressing the *gambiense*-VSGs LiTat 1.3 or LiTat 1.5 be effectively used as a safe and reliable alternative for diagnosing HAT while maintaining high specificity and enabling high-throughput detection in ITL?

Hypothesis: We hypothesise that the *in vitro* cultivated transgenic *T. b. brucei* strains, expressing the LiTat 1.3 or LiTat 1.5, will demonstrate comparable diagnostic performance to the *T. b. gambiense* strains in the ITL. Furthermore, incorporating fluorescent reporter genes TagGFP2 or DsRed into the transgenic line can improve the ITL process, allowing for easier and more rapid multiplex detection using the Spark CYTO 400 (Tecan). We expect that using these transgenic strains, combined with fluorescence-based detection, will enhance the throughput while maintaining the required specificity, ultimately aiding in the post-elimination phase of gHAT.

The different aspects of the experimental work in this thesis are summarised in Figure 7. The first part will validate whether the existing transgenic *T. b. brucei* strains solely express the *gambiense*-specific VSGs LiTat 1.3 or LiTat 1.5. Therefore, these strains will be subjected to an ITL with antibodies specific to these VSGs to ensure that the strains can be used in subsequent experiments. Focussing on the biohazard risk, this part will also check whether these *in vitro* cultivated strains have the potential to replace the *gambiense*-strains that are now used in the ITL. On the one hand, this would prevent the risk of infection of laboratory personnel; on the other hand, it reduces the number of laboratory animals, thereby complying with the 3Rs (replacement, reduction, and refinement). If the *T. b. brucei* strains are found to be effective in the ITL, this will allow us to test the antibody-mediated complement lysis using a non-human infective species that can be propagated *in vitro*, making the test safer and animal-friendly. However, it is possible that a way to avoid the lysis of *T. b. brucei* by ApoL1 should be found.

In the final part, we will focus on improving the ITL. Fluorescent reporter genes will be expressed in this transgenic line, allowing easier and more rapid multiplex detection using the Spark CYTO 400. Finally, to determine the diagnostic capacity, the performance of the new ITL will be compared to the original one in a concordance analysis and validated against a batch of human sera.

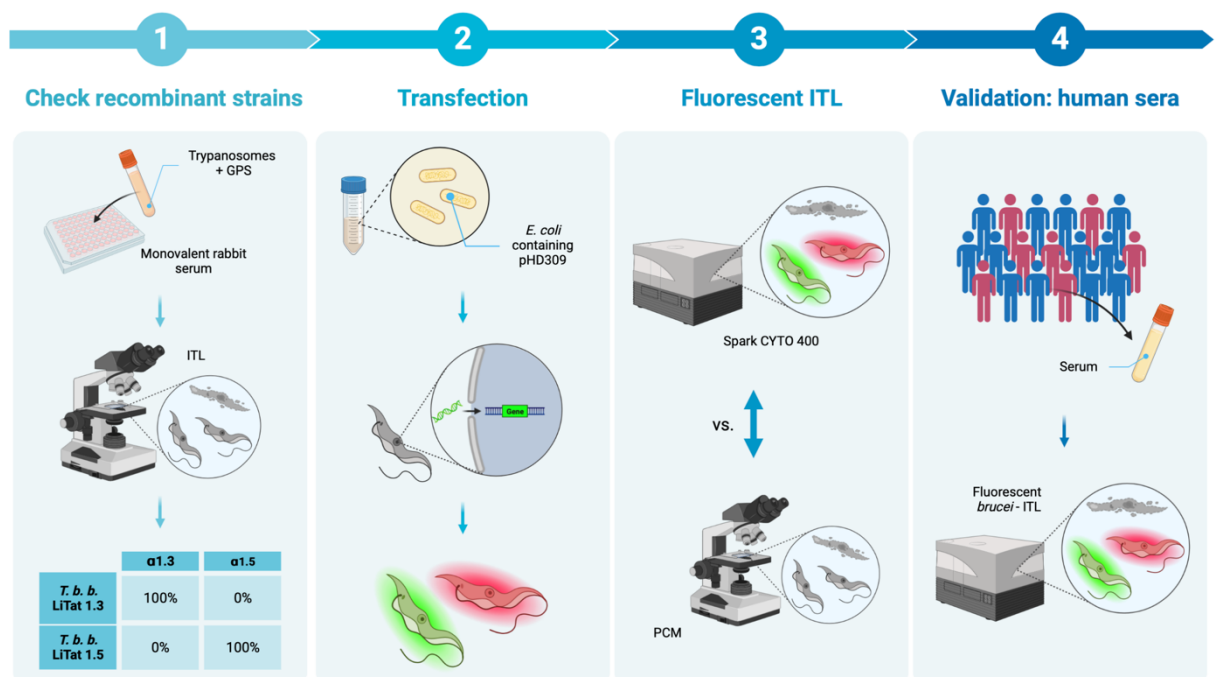


Figure 7: Graphical abstract of the workflow depicting the experimental steps.

3. Materials and Methods

3.1 Parasite strains

The used strains are described in Table 1. The research mainly focused on the earlier described *T. b. brucei* Lister427 strains expressing either LiTat 1.3 or LiTat 1.5 (Figure 6). The antibiotic resistance genes for puromycin and blasticidin prevent the switching to other VSGs in the presence of the mentioned selection antibiotics. To

optimise the experimental protocol of the Spark CYTO 400 while the *T. b. brucei* Lister427 strains were tested and grown, two *T. b. brucei* Antwerp Trypanozoon antigen Repertoire (AnTaR) 1 strains were used that express either TagGFP2 or DsRed. The pH309 vector was used to integrate the constitutively expressed fluorescent reporter genes into the β -tubulin region of these strains [64, 65]. Additionally, *T. b. gambiense* clones were used for the infection of mice and, subsequently, ITL.

Table 1: *T. b. brucei* strains used in the experiments.

Strain	Variant	Vector	Antibiotics	Culture method
<i>T. b. brucei</i> Lister427	LiTat 1.3-SE	pBS-SK ⁺	puromycin + blasticidin	<i>in vitro</i> , <i>in vivo</i>
	LiTat 1.5-SE	pBS-SK ⁺	puromycin + blasticidin	<i>in vitro</i> , <i>in vivo</i>
<i>T. b. brucei</i> AnTaR 1	TagGFP2	pHD309	hygromycin	<i>in vitro</i>
	DsRed	pHD309	hygromycin	<i>in vitro</i>
<i>T. b. gambiense</i>	LiTat 1.3	N/A	N/A	<i>in vivo</i>
	LiTat 1.5	N/A	N/A	<i>in vivo</i>

3.2 Culture methods

3.2.1 *In vivo* culture in OF1 mice

The care and experimental procedures involving mice were conducted in accordance with ethical guidelines approved by the Animal Ethical Committee of the Institute of Tropical Medicine (No. TU2021-1).

For the *in vivo* method, female OF1 mice were intraperitoneally infected with the *T. brucei* strains. A suspension starting from 2×10^5 parasites in a maximum volume of 200 μ L was made and injected. When the strains for these injections were derived from *in vitro* cultures, the selection antibiotics were removed by centrifuging the culture at 1500 relative centrifugal force (rcf). The supernatant was discarded and replaced with phosphate-buffered saline + 1% glucose pH 8.0 (PSG). Alternatively, blood from one mouse was collected, diluted with PSG if necessary and injected into a recipient mouse. After 48 hours past infection, the mice's parasitaemia was checked using the matching method (MM)[66]. This involved taking 5 μ L blood from a tail-snip and estimating the number of parasites per field when placed under a 24×24 mm coverslip (10×40 magnification, PCM). At day 3 or 4 past infection, when parasitaemia reached antilog 8.1–8.4 in at least one mouse, the mice were euthanised using euthasol (pentobarbital) at a dose of 0.1 mL/100 g. This specific time window ensures that the majority of the parasites still express the desired LiTat variant. The blood was collected via cardiac puncture after the syringe was rinsed with a drop of heparin. The blood was passed over a mAECT column (max. 0.4 mL/column) at pH 8 to separate the trypanosomes from the blood and suspend them in a minimum volume of 3 mL PSG. These trypanosomes were then put in *in vitro* culture or stored as pellets. For the pellet, the elution was centrifugated, the supernatant was removed, and the sedimented trypanosomes were stored for later use as frozen pellets at -80°C . Alternatively, when blood from cardiac puncture was mixed with 10% glycerol, the infected blood could also be used to make glycerol cryostabilates, which are cryopreserved stocks of the *T. b. brucei* strains. Glycerol cryostabilates serve as a method of long-term storage for the strains, ensuring the viability and integrity of the strains and allowing for consistent and reliable cultures to be established from the cryostabilates.

3.2.2 *In vitro* culture in HMI-9 medium

The *in vitro* cultures started from glycerol cryostabilates. The different *T. b. brucei* strains (Table 1) were grown in Hirumi's Modified Iscove's medium-9 (HMI-9) medium [67] with selection antibiotics (if applicable). For the preparation of HMI-9 medium, Iscove's Modified Dulbecco's Medium (IMDM, Gibco™) with NaHCO_3 was supplemented with 100x stock solutions of 100mM hypoxanthine, 5 mM bathocuproine disulfonate, 20 mM β -mercaptoethanol, 16 mM thymidine, 100 mM sodium pyruvate and 100 mM L-cysteine (all from Sigma-Aldrich). Additionally, 20% of heat-inactivated horse serum (Gibco™) was added together with the selection antibiotics

puromycin (0.2 µg/mL) and blasticidin (5 µg/mL) from InvivoGen to avoid switching of the VSGs. Finally, the solution was sterilised over a 0.22 µm bottle top filter and stored at 4°C.

Four serial tenfold dilutions of the cultures were made in prewarmed medium and incubated at 5% CO₂ and 37°C in a 24-well plate in 500 µL. After 48-72 hours, the trypanosomes reached a concentration of around 10⁶ cells/mL. The populations' morphology and motility were checked using an inverted phase-contrast microscope and counted using disposable counting chambers (Uriglass, Menarini). Further dilutions of the best-fit population were made down to 10⁵ cells/mL to keep them in their logarithmic growth. When needed, the cultures were progressively scaled up in culture flasks (25 cm²) from 500 µL to 4 mL on the first day. Subsequently, the culture volume was further increased from 4 mL to 40 mL on the second day. Finally, a last expansion expanded the culture to a volume of 400 mL in 150 cm² flasks to obtain enough trypanosomes for testing or transfection.

For long-term storage, the culture volume was centrifuged at 1500 rcf and 4°C. The supernatant was removed, and the pellet was concentrated tenfold in HMI-9 with 10% glycerol. Aliquots of 1 mL were then frozen using cryogenic freezing equipment (NICOOOL FREEZAL, Air Liquide) and stored in liquid nitrogen.

A unified protocol for sample preparation and staining was implemented. The general steps involved: the concentration of the *in vitro* culture was calculated, and the appropriate amount of culture was centrifuged at 1500 rcf to obtain sufficient cells. The cells were then washed in PSG and centrifuged again. Depending on the protocol, the trypanosomes were resuspended in either PSG, guinea pig serum (GPS) or water (to simulate dead cells through hypotonic lysis). In cases where staining was required, 50 µL of PSG containing the desired stain was added to the wells or the cells were incubated with the stain for 20 min. The staining solutions used included Hoechst in a final concentration of 5 µg/mL, propidium iodide (PI) in a final concentration of 1 µg/mL, or a combination of both.

3.3 Microscopy

Cytospin 4 (Thermo Scientific) centrifugation and Giemsa staining observations were performed to visualise the morphology of the trypanosomes. First, trypanosomes were collected from the *in vitro* cultures at a minimum concentration of 0.5 × 10⁶ cells/mL in HMI-9. Next, a 200 µL suspension was loaded into the cytofunnel and centrifuged at 1000 g for 5 min. The air-dried cytospin slides were then fixed in methanol for 5 min and stained with Giemsa (1/10 dilution in phosphate buffer pH 7.0) for 20 min. Then, the Giemsa slides were examined using light microscopy. Additionally, trypanosome slides were prepared directly from the cultures and subjected to inspection using PCM (Leica DM2000 LED) after concentration through centrifugation. Images were captured using the Leica Application Suite V4 software.

3.4 Epifluorescence microscopy

The fluorescent trypanosomes were also visualised with epifluorescence microscopy (Olympus BX41). Similarly to conventional microscopy, *in vitro* cultures were concentrated. When the visualisation of Hoechst 33342 (Invitrogen) was preferred, 5 µg/mL was added. The *T. b. brucei* were observed with the use of the DAPI, GFP and TRITC filter for blue, green, and red fluorescence, respectively. The Olympus cellSens Standard software was used for the acquisition of images and allowed the formation of overlay images and background correction.

3.5 Immune trypanolysis

The antisera were obtained from the collection of the Applied Technology and Production Unit of ITM, initially established by Van Meirvenne [68]. These antisera originate from VAT clones (isolated from a patient in Ivory Coast in 1952 [69, 70]) that were inoculated in rabbits, subsequently aliquoted and freeze-dried for long-term storage. Each aliquot was labelled based on the rabbit's identifier and number of days post-infection (dpi) after which the serum was obtained (e.g., K446/7 indicates rabbit No. 446, seven dpi). Before use, each aliquot was solubilised in 200 µL of water.

For the ITL, 25 μ L serum was mixed with 25 μ L GPS (Envigo) in a 96-well plate. The plate was placed on a microplate shaker and incubated for 30 min at room temperature (RT). Afterwards, 50 μ L trypanosome suspension in GPS from the *in vivo* or *in vitro* culture was added at 5-10 trypanosomes per field (10 $\times$ 40 magnification, PCM). After 90 min incubation on a microplate shaker at RT, 8 μ L of each well was placed on a slide and covered with an 18 $\times$ 18 mm coverslip. The test was considered positive if more than 50% of the trypanosomes were lysed after observing a minimum of ten fields under the microscope (10 $\times$ 25 magnification, PCM).

3.6 Transfection

3.6.1 *E. coli* culture and growing of plasmids

For the stable integration of fluorescent reporter genes into the genome of the *T. b brucei* LiTat 1.3 and LiTat 1.5 strains, we used the pHD309-plasmid [64] (Supporting figure 2; Supporting figure 3). This plasmid already contains either the DsRed (red fluorescence) or TagGFP2 (green fluorescence) genes and a hygromycin-resistance gene that can be used for the selection of recombinant trypanosomes. First, 5 μ L of *E. coli* bacteria containing the plasmid were grown overnight at 37°C in Luria-Bertani (LB) broth containing the selection antibiotic carbenicillin (0.1 mg/mL). Next, the QIAprep Spin Miniprep Kit (QIAGEN) was used according to the manufacturer's protocol to isolate the plasmids from the bacteria cultures. Finally, the concentration of the extracted plasmids was measured using the NanoDrop One UV-Vis spectrophotometer.

3.6.2 Restriction digestion

After plasmid purification, restriction digestion was performed using the enzymes NotI-HF or BamHI and HindIII-HF and the products were purified using the QIAquick PCR Purification Kit (QIAGEN). Ethanol precipitation was conducted to concentrate the obtained DNA by adding 1/10 volume of 3 M Na-acetate and 2.5 volumes of 100% ethanol to the DNA. The tubes were then incubated at -20 °C overnight. The solution was centrifuged at high speed for 30 min at 4°C, forming a DNA pellet. The supernatant was removed, and the tubes were opened to drain any remaining liquid. Subsequently, the pellet was washed with 500 μ L of cold 70% ethanol to remove excess salt, followed by another round of centrifugation at high speed for 5 min at RT. The supernatant was discarded, and the pellet was left to dry. The dried DNA was then concentrated in water in a volume up to 20 times smaller than initially.

3.6.3 Gel electrophoresis

In order to validate the restriction digestion, we performed gel electrophoresis on a 0.8% agarose gel in TAE buffer. The samples were mixed with loading dye and loaded onto the gel. Different DNA ladders were included as size markers. The gel was run at a constant voltage of 135V for 35 min. Following electrophoresis, the gel was stained with ethidium bromide, and the DNA bands were visualised under UV light.

3.6.4 Transfection into the β -tubulin region

For the transfection, we started from both *in vivo* and *in vitro* cultures. The TagGFP2 and DsRed genes were integrated into the β -tubulin repeats (Supporting figure 4) and transfected using the Amaxa Basic Parasite Nucleofector Kit 2. Briefly, we centrifuged the required cells ($\pm 5 \times 10^7$) and replaced the supernatant with 100 μ L of Nucleofector Solution per sample. Next, 5-10 μ g DNA in 1-5 μ L was added to the cell suspension, and the suspension was transferred to the cuvettes, after which the U-033, X-001 or D-023 program was run. After electroporation, the transfected cells were immediately transferred to prewarmed HMI-9 medium (0.2 μ g/mL puromycin and 5 μ g/mL blasticidin) to allow recovery. The trypanosomes were then seeded on a 48-well plate in 250 μ L and incubated overnight. The next day, 250 μ L of the HMI-9 containing the selection antibiotic hygromycin

B gold (20 µg/mL) was added to the wells and the plate was further incubated. After a sufficient duration of antibiotic selection, the wells were checked for survival, growth, and expression of the gene of interest.

3.7 Spark CYTO 400

3.7.1 Fluorescence intensity bottom reading

Fluorescence intensity (FI) was measured using the Spark CYTO 400 instrument. The FI module of the instrument uses a monochromator that enables the precise selection of a specific wavelength from the optical spectrum. Specific parameters were defined for different fluorophores to set up the fluorescence intensity measurement. For TagGFP2-labelled trypanosomes, an excitation wavelength of 487 nm with a bandwidth of 20 nm was selected, while the emission wavelength was set at 530 nm with a bandwidth of 50 nm. Similarly, for DsRed-labelled trypanosomes, an excitation wavelength of 545 nm with a bandwidth of 20 nm was chosen, and the emission wavelength was set at 585 nm with a bandwidth of 50 nm. These specific combinations of excitation/emission wavelengths and bandwidths were determined to maximise excitation efficiency while minimising interference from background signals or crosstalk.

3.7.2 Fluorescence imaging mode

The fluorescence imaging mode of the Spark CYTO 400 instrument was used to count the fluorescently labelled trypanosomes. Both built-in 10x and 4x objectives of the Spark were used. In the SparkControl software, the blue channel was configured with the following settings: LED intensity of 100%, focus offset of 0 µm, and sensitivity of 95%. Object sizes for the blue channel were set with a minimum length and width of 4 µm and a maximum length and width of 15 µm. Similarly, for the green and red channels, the settings for object sizes were as follows: minimum and maximum object lengths of 4 µm and 40 µm, respectively, and minimum and maximum object widths of 4 µm and 20 µm, respectively. A sensitivity of 95% was chosen for both channels. Due to the nature of the imaging process, the exposure time needed to be determined individually for each channel during the initial setup using the auto-exposure function in the Live Viewer. Unlike exposure time, the sensitivity and object sizes are parameters that can be adjusted in the ImageAnalyzer software, even after the initial image acquisition. The obtained images were analysed based on the predefined settings for each channel. The ImageAnalyzer software uses an algorithm specifically designed to detect individual objects within an image. Therefore, the software can provide results such as fluorescence per object, object size, and the total number of objects detected.

As part of the experimental procedure, an initial shaking step was consistently included before the image acquisition. This step aimed to ensure proper mixing and distribution of the sample components. Subjecting the sample to a shaking motion could disrupt any sedimentation, allowing for a more homogeneous sample. This involved utilising a double orbital shaking mode of 30 seconds, an amplitude of 1 mm, and a frequency of 270 revolutions per minute (rpm). Following the shaking step, a brief waiting period of 20 seconds ensured that the fluid in the well had a chance to stabilise.

In addition, for kinetic experiments (capturing a series of images over a defined period, allowing the observation of dynamic changes within the sample) conducted with the Spark, the same protocol was repeated for a specific number of cycles.

3.7.2.1 Dilution series

A ½ dilution in 100 µL was made in a black 96-well plate (Corning) with a starting concentration between 1.5×10^6 and 3.2×10^6 parasites per well. This concentration is equivalent to 8 and 16 trypanosomes per field, respectively, thus similar to the concentration used in the ITL (i.e., 5-10 per field). This was done to facilitate the comparison with the original ITL. The Spark was run using the 4x and 10x objective and the appropriate settings.

3.7.2.2 Kinetic *in vitro* ITL

All human samples for the validation in this study were collected within the framework of the HAT-PolyB diagnostic study and comprised samples of which the ITL result was already known [71]. Only a subset of samples from the HAT-PolyB study was used to achieve a balanced distribution of controls and patients for analysis. In addition, specific efforts were made to include samples that tested positive for only one of the LiTat variants (LiTat 1.3 or 1.5) to demonstrate the proof-of-principle.

The cells were prepared as described in section 3.2.2 to obtain a concentration of 7.5×10^5 per well for the AnTaR 1 strains and 4×10^5 per well for the LiTat strains. A black 96-well plate (Greiner) was prepared by adding 25 μ L of serum and 25 μ L of GPS to each well. The cells were distributed evenly over the different wells, and the plate was transferred to the Spark reader. A kinetic fluorescence imaging loop was initiated for data acquisition using the 4x (one imaging field) or 10x objective (three imaging fields, Supporting figure 6).

3.8 Statistical analysis

The data obtained from the Spark CYTO 400 instrument was analysed in the Spark's ImageAnalyzer software. Raw data obtained from the Spark instrument was retrieved in Excel. Initial organisation and processing, including necessary data transformations, such as normalisation and standard deviations (SD), were performed within Excel. In order to visualise the experimental results and trends, graphs were generated using the R statistical software.

To assess the degree of agreement between the different ITL methods, Cohen's kappa coefficient was calculated using the formula:

$$\kappa = \frac{P_o - P_c}{1 - P_c}$$

where P_o represents the proportion of observed agreement, and P_c is the proportion of agreements expected by chance. Kappa values below 0 reflect a poor strength of agreement, while values from 0-0.20 reflect 'slight', 0.21-0.4 'fair', 0.41-0.60 'moderate', 0.61-0.8 'substantial', and above 0.81 'almost perfect' strengths [72, 73].

4. Results

4.1 Cultivation of the *T. b. brucei* strains

In comparison to the *T. b. brucei* AnTaR 1 cultures, the transgenic *T. b. brucei* Lister427 LiTat 1.3 strain exhibited a bloated and deformed appearance compared to the normal morphology (especially when reaching the mid-log phase) when observed via PCM. A Giemsa slide was made using the cytopspin system to observe the distinct morphology in more detail. The staining further underlined the morphological abnormalities (Figure 8). Furthermore, the *T. b. brucei* LiTat 1.3 presented a slower growth rate across the multiple passages than the LiTat 1.5 strain. These malformations could be related to the transgenic modifications that integrated the LiTat 1.3 and antibiotic-resistance genes into the genome. The slower growth and remarkable morphology suggest that, in the case of LiTat 1.3, the transgenic modifications significantly impacted its morphology, which in turn affects its growth.

Due to the slow growth and abnormal morphology, we passed the strain through mice in an attempt to re-activate their growth. After three consecutive *in vivo* passages, we observed a significant improvement in the *in vitro* growth rates when re-entered in culture medium. Indeed, after the passages, the strain displayed an increased replication rate and reached higher cell densities within the same culturing period. Moreover, we

observed that a smaller portion of the trypanosomes exhibited malformations, although the atypical morphology did not completely disappear after the *in vivo* passages.

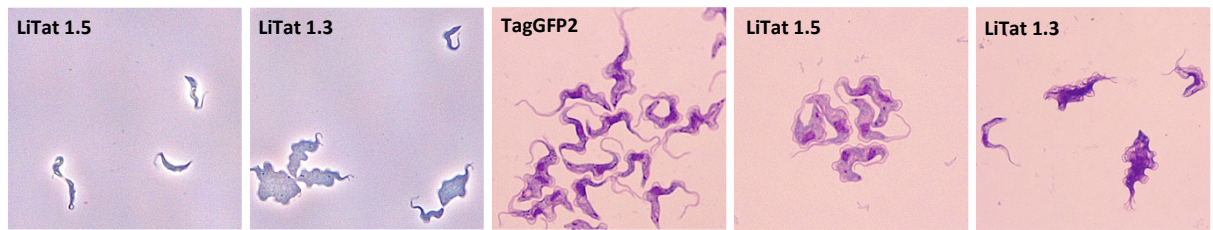


Figure 8: Microscopy images of the *in vitro* cultures.

The images represent examples of the observed morphologies. The first two pictures (transgenic *T. b. brucei*) show PCM images captured with a 40x objective. The last three images are captured fields of the cyospin slides (100x objective). The pictures show the typical morphology of the BSF strains, including the abnormal morphology of the *T. b. brucei* LiTat 1.3 strain in comparison to the AnTaR 1 and LiTat 1.5 strains.

4.2 ITL of the *T. b. brucei* LiTat 1.3 and LiTat 1.5 strains

Intending to validate the correct expression of the *gambiense*-VSG on the surface of the transgenic *T. b. brucei* strains, we investigated the reactivity of the transgenic strains compared to their *T. b. gambiense* counterparts using the immune trypanolysis assay. A panel of specific *T. b. gambiense* LiTat-variant rabbit antisera, obtained after a well-defined number of days post-infection (dpi) were used. Additionally, we investigated the effect of different culturing methods on the lysis patterns of the transgenic LiTat strains. The results of the immune trypanolysis test in Table 2 confirm the successful expression of the LiTat 1.3 and 1.5 by the parasites. Overall, the strains were only lysed by their specific monovalent antisera or sera that could be considered polyvalent. We also included three different NHS to see whether the lytic effect of ApoL1 could have an impact on the results of the ITL. Based on the observed results, the NHS did not seem to have an interfering trypanolytic activity on the *T. b. brucei* strains after 90 min. While the transgenic strains showed similar reactivity when cultured *in vitro*, a difference was observed after four passages (P4) in mice. Specifically, the *T. b. brucei* LiTat 1.3 was lysed by sera against the VSGs 117 and 221. This observation suggested that the absence of selection antibiotics in the *in vivo* environment allowed the transgenic strains to undergo VSG switching, a phenomenon known to be associated with immune evasion.

4.3 Explorative experiments using the Spark CYTO 400

4.3.1 Reading mode evaluation

Our preliminary experiments allowed us to evaluate the Spark's performance in detecting the fluorescent signal from the AnTaR 1 strains. The fluorimetry experiment using the FI mode showed similar relative fluorescence units (RFU) for both green, red, and unlabelled parasites (Figure 9). Moreover, when a dilution series in PSG was made, no change in RFU values was seen with decreasing concentration of cells. Therefore, the FI mode was excluded from further testing.

With the fluorescence imaging mode of the Spark CYTO 400, a linear relationship was seen between the number of trypanosomes per well and the number of objects counted by the machine. With the Spark's sensitivity settings set to 95% and object sizes between 4 μm and 40 μm , the data demonstrated that the number of fluorescent counts increased linearly as the number of trypanosomes per well increased (Figure 10). However, this relationship reached a plateau between 7.5×10^5 and 1.5×10^6 cells when using the 10x objective. This number of cells aligns with approximately 4 or 8 per field, respectively, when using the MM (Supporting table 1). This suggests that the ideal concentration of trypanosomes for the Spark CYTO 400 corresponds to that used in the ITL.

Table 2: Results of the immune trypanolysis test against different sera.

The strains obtained via *in vivo* and *in vitro* culture were tested against different rabbit sera to prove they expressed the correct VSGs. The transgenic strains were also compared to the standard *T. b. gambiense* strains. The results show the percentage of trypanolysis (i.e., 100% = all trypanosomes were lysed or dead, 0% = all trypanosomes were alive after 90 min of incubation.) compared to the negative controls GPS and PSG. We used lyophilised sera collected after a certain number of days past infection (indicated by the number after the forward slash "/") with the mentioned strain. The number of the passage of the *in vivo* culture is indicated by 'P'.

Serum \ Strain	<i>T. b. gambiense</i> LiTat 1.3 (<i>in vivo</i>)	<i>T. b. brucei</i> LiTat 1.3 (<i>in vivo</i> , P1)	<i>T. b. brucei</i> LiTat 1.3 (<i>in vivo</i> , P2)	<i>T. b. brucei</i> LiTat 1.3 (<i>in vivo</i> , P4)	<i>T. b. gambiense</i> LiTat 1.5 (<i>in vivo</i>)	<i>T. b. brucei</i> LiTat 1.5 (<i>in vivo</i> , P1)	<i>T. b. brucei</i> LiTat 1.5 (<i>in vitro</i>)
GPS	0	0	0	0	0	0	0
PSG	0	0	0	0	0	0	0
α-VSG 117	0	0	0	<u>100</u>	0	0	0
α-VSG 221	0	0	10	<u>80</u>	0	0	0
NRS	0	0	0	0	0	0	0
R-008 AnTat1.1/26	<u>60</u>	<u>80</u>	<u>60</u>	<u>80</u>	<u>60</u>	<u>70</u>	<u>90</u>
NHS-ND	0	0	0	0	0	0	0
NHS-NVR	0	0	0	0	0	0	0
NHS-PB	0	0	0	0	0	0	0
LiTat 1.1 K446/7	0	0	0	0	0	0	0
LiTat 1.2 K478/6	0	0	0	0	0	0	0
LiTat 1.3 K551/9	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	0	0	0
LiTat 1.3 K551/12	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>
LiTat 1.3 K552/9	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	0	0	20
LiTat 1.3 K552/12	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>
LiTat 1.3 K685/28	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>
LiTat 1.4 K372/6	0	0	0	0	0	0	0
LiTat 1.5 K377/6	0	0	0	0	<u>100</u>	<u>100</u>	<u>100</u>
LiTat 1.5 K378/14	<u>100</u>	<u>100</u>	<u>60</u>	<u>60</u>	<u>100</u>	<u>100</u>	<u>100</u>
LiTat 1.5 K378/120	<u>100</u>	<u>100</u>	<u>80</u>	<u>90</u>	<u>90</u>	<u>100</u>	<u>100</u>
LiTat 1.6 K425/4	0	0	0	0	0	0	0
LiTat 1.6 K425/5	0	0	0	0	0	0	0
LiTat 1.6 K425/8	0	0	0	0	0	0	0
LiTat 1.6 K425/20	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>	<u>100</u>
LiTat 1.7 K463/6	0	0	0	0	0	0	0
LiTat 1.8 K465/6	0	0	0	0	0	0	0
LiTat 1.9 K466/6	0	0	0	0	0	0	0
LiTat 1.10 K604/6	0	20	20	30	0	0	20

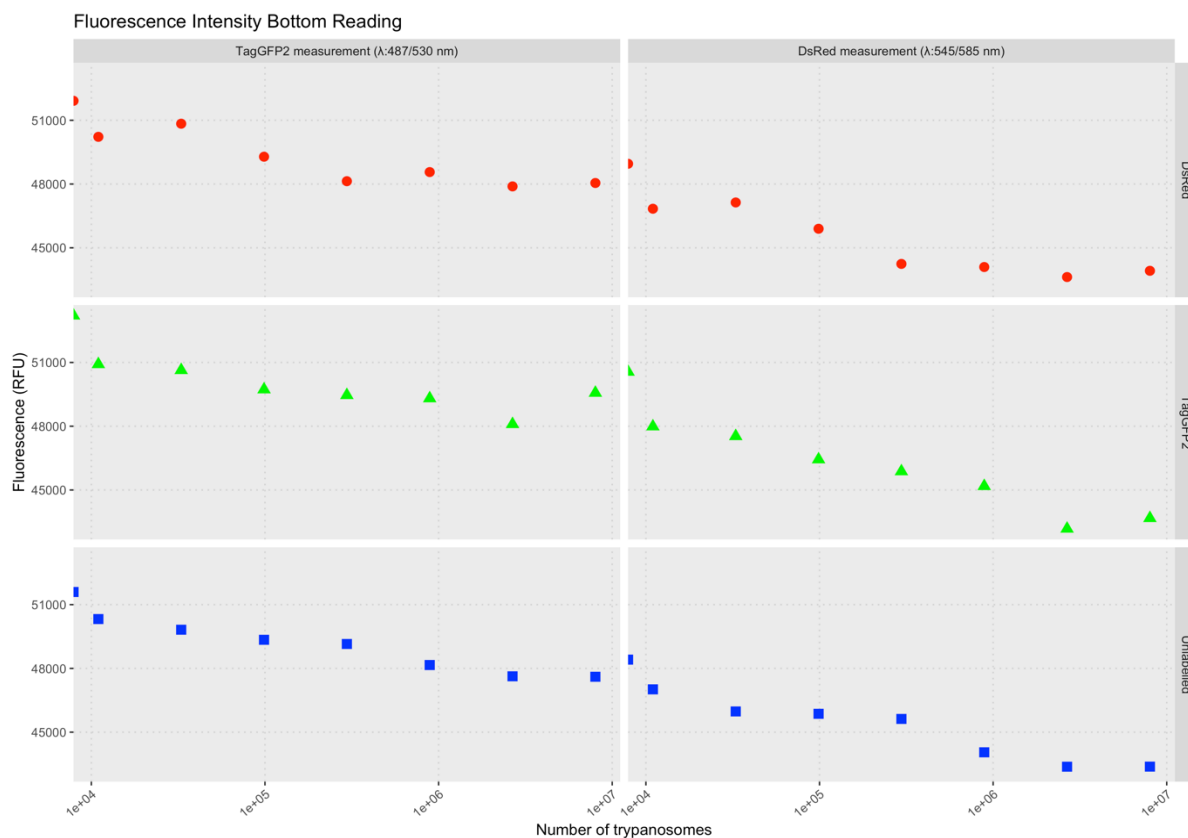


Figure 9: Fluorimetry experiment using the Spark CYTO 400.

The number of cells plotted against the RFU in two different channels using a dilution series of *T. b. brucei* AnTaR 1 TagGFP2, AnTaR 1 DsRed and unlabelled *T. b. brucei* LiTat 1.5 strains.

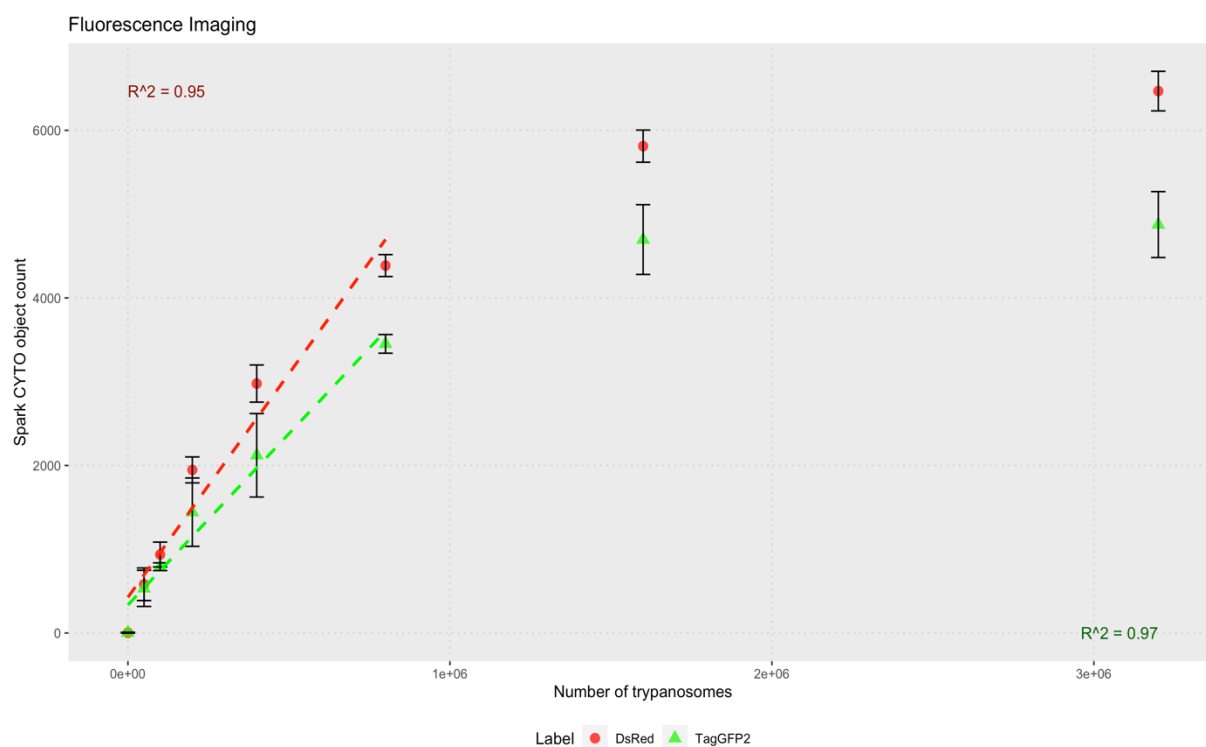


Figure 10: Relationship between the number of trypanosomes and object counts by the Spark CYTO 400.

The graph shows the number of counts obtained by the Spark (10x objective) in relation to an increasing number of trypanosomes (AnTaR 1) per well. Trendlines are added to the linear part of the plot with indication of their R^2 value. Error bars indicate SD.

4.3.2 DNA staining and lysis experiment

The effect of adding Hoechst 33342 and PI in the Spark experiment was investigated. We used different concentrations of the *in vitro* cultivated AnTaR 1 TagGFP2 trypanosomes and compared the number of counts obtained via the Spark between the fluorescent label and the Hoechst and/or PI staining. As shown in Figure 11, adding Hoechst produced nearly identical object counts as when using the TagGFP2 label alone, indicating that Hoechst is a reliable trypanosome stain. However, when PI was included, we saw that the Spark reader detected green objects, even though the cells were completely lysed beforehand (Figure 11, bottom right). Indicating that PI causes severe crosstalk in the other emission channels of the reader, thus generating false-positive green object counts. Adapting the Spark's crosstalk correction settings failed to resolve this issue sufficiently, even with other PI concentrations. Therefore, we decided to exclude the addition of PI in the other experiments.

Besides the staining options, we also examined the difference between the 4x and 10x objectives (Supporting figure 5 shows the results for the 10x objective). Since the 4x objective could detect and consequently count the trypanosomes, we preferred to use this objective as it can capture a larger field of the well. The larger area is less prone to inadequate counts when events such as agglutination cause the cells not to be evenly distributed over the well.

We determined the optimal settings and experimental protocol for detecting the fluorescent signals based on the beforementioned results. We opted to include Hoechst as an extra validation method and decided to work further with a single concentration of 7.5×10^5 trypanosomes per 50 μ L. This concentration is high enough to avoid a significant effect of unspecific background counts but low enough to prevent reaching the point of saturation.

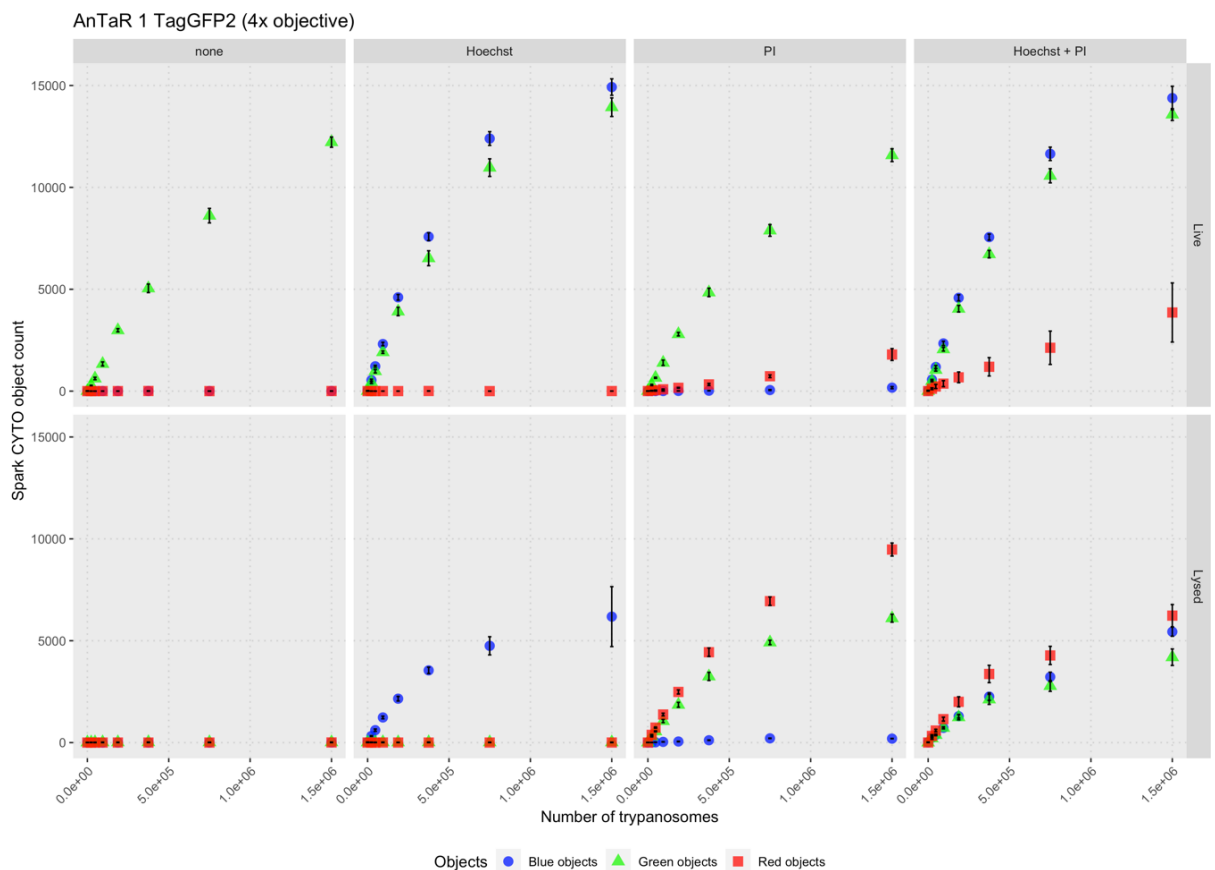


Figure 11: The addition of Hoechst and PI to live and lysed AnTaR 1 TagGFP2 cells (green).

For the reading in the Spark, cells were either live or lysed by hypotonicity. The graph shows the objects counted by the Spark for the following conditions: no stain added, addition of Hoechst (blue), PI (red), or a combination of both stains. Error bars indicate SD.

4.3.3 Immune trypanolysis

The trypanolytic effect of sera on the *T. b. brucei* AnTaR 1 strains was evaluated in a kinetic experiment. The trypanosomes were either resuspended in PSG or GPS to investigate the effect on kinetics of the trypanolysis assay. Fluorescence readings were recorded over a period of approximately 150 minutes using the 4x objective. The addition of the antiserum R-008 (AnTat 1.1, 26 dpi) led to a notable decrease in parasites over time (Figure 12). This effect was consistent in all the replicates and the same trend was seen in both the TagGFP2 and DsRed strains, suggesting that the Spark is indeed capable of monitoring the lysis of the trypanosomes by antisera in a kinetic measurement. Furthermore, the reaction was slightly slower when the trypanosomes were resuspended in PSG, suggesting that the choice of resuspension buffer may affect the kinetic profile of the assay. This is likely due to the lower amount of complement present in the well. Strikingly, the lytic effect of ApoL1 in NHS did not seem to have a pronounced effect, as the counts remained relatively stable during the course of the experiment. This aligns with what we observed during the conventional ITL experiment, indicating that TLFs might not interfere with the ITL protocol. In addition to the main observations of the kinetic experiment, another notable finding was the decrease in blue object counts (stained with Hoechst) throughout the experiment. It is worth noting that, in the case of the PSG control, the green and red fluorescence measurements of the trypanosomes remained relatively stable over time, indicating that the observed decrease in blue object counts was not due to a loss of viability.

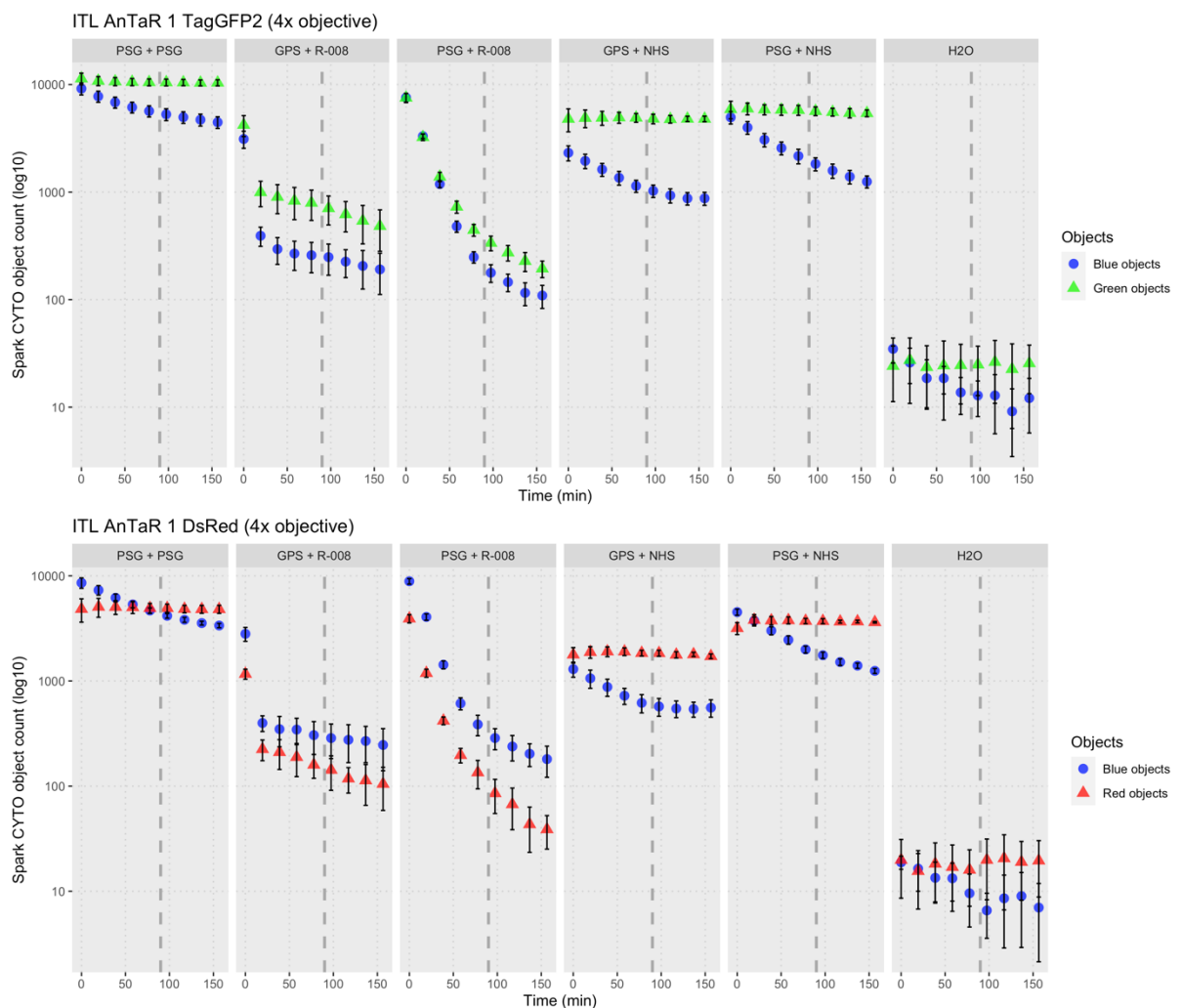


Figure 12: Kinetic course of the trypanolytic action of different sera detected by Spark CYTO 400.

The fluorescent AnTaR 1 trypanosomes were suspended in either PSG or GPS and incubated with Hoechst. After the addition of GPS and either PSG, NHS or R-008 serum the plate was read with the Spark. The hypotonic lysis of water was added as control. The dashed line shows the endpoint of the normal ITL (i.e., 90 min). Error bars indicate SD. For both fluorescent labels, similar results were obtained.

In addition to presenting the raw data, we also utilised normalised data in our analysis to provide a clearer perspective on the experimental results (Figure 13). The trypanosome counts were therefore normalised by using their initial value as the baseline. Since the object counts at the start of the measurement differ between the conditions, the normalisation allowed the comparison and interpretation of data by scaling the counts to their own starting point. This allowed us to assess relative changes in fluorescent object counts over time or between different experimental conditions.

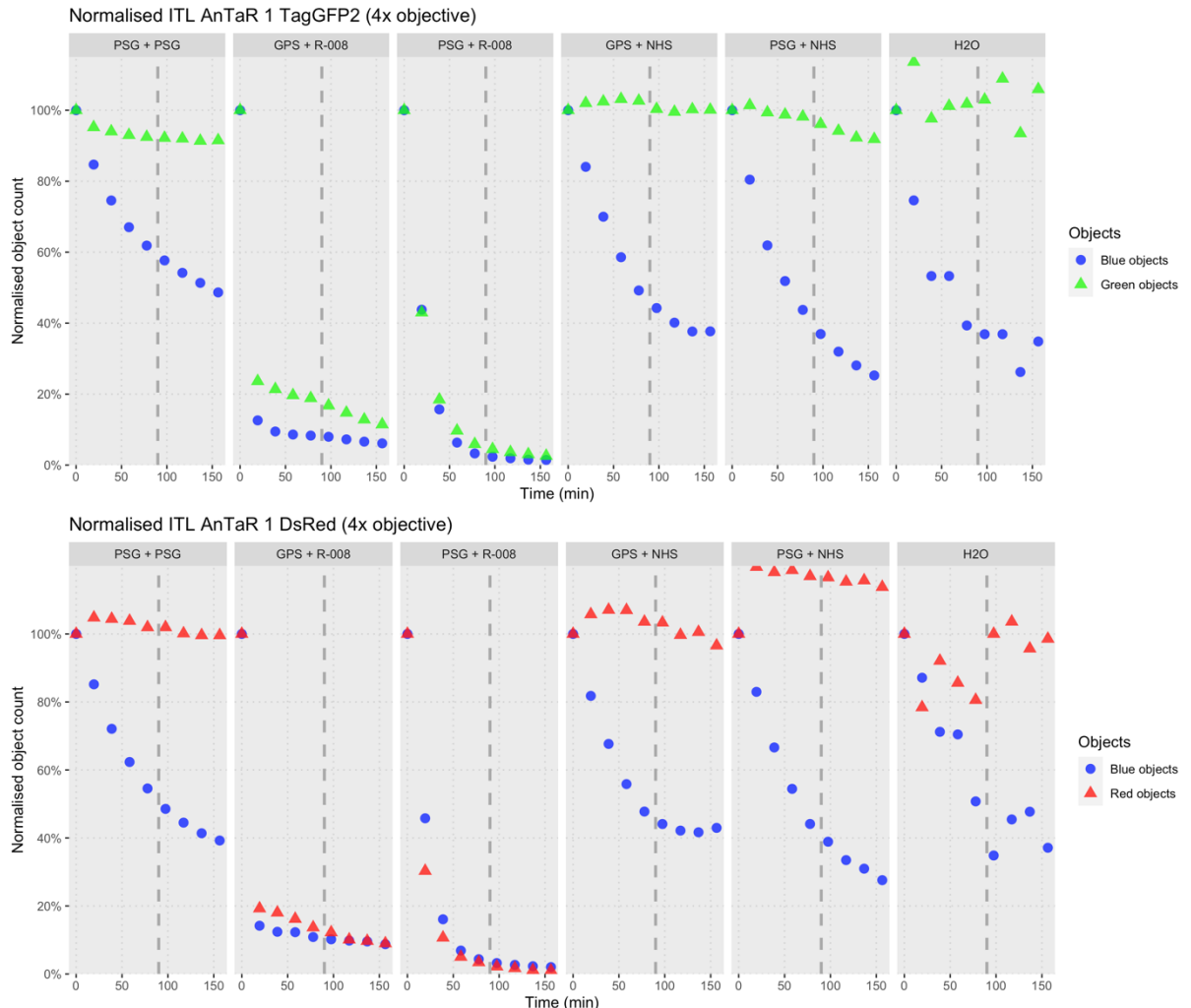


Figure 13: Kinetic course of the trypanolytic action of different sera (normalised).

The graph shows the same results as Figure 12. However, the object counts were normalised by dividing the counts for each condition by their own starting value. The dashed line shows the endpoint of the normal ITL (i.e., 90 min).

4.4 Generation of fluorescent *T. b. brucei* LiTat 1.3 and LiTat 1.5

4.4.1 Plasmid purification and linearisation

After the plasmids were purified, restriction digestion was performed using NotI-HF or BamHI and HindIII-HF. Following QIAquick PCR purification, the resulting products were loaded on an agarose gel for electrophoresis to validate the digestion with the restriction enzymes. Upon inspection, distinct bands could be observed on the gel (Figure 14). The band sizes were compared to the expected sizes based on the known sequences. The slots loaded with the TagGFP2 and DsRed plasmids cut using NotI-HF showed a band at around 6500 bp. Additionally, where the restriction enzymes BamHI and HindIII-HF were used, bands at approximately 6000 bp and 700 bp were seen. These bands correspond to the expected sizes of the different TagGFP2 pHD309 fragments: 6259 bp for the NotI-HF restriction digestion and 5536 bp and 723 bp for the BamHI and HindIII-HF cuttings sites. Similarly, the restriction digestion of DsRed pHD309 revealed bands related to the fragments of 6220 bp (NotI-HF) and both

5536 bp and 684 bp (BamHI - HindIII-HF). Therefore, the gel electrophoresis showed the successful digestion of the plasmids and thus ensured that the fragments could be used for transfection.

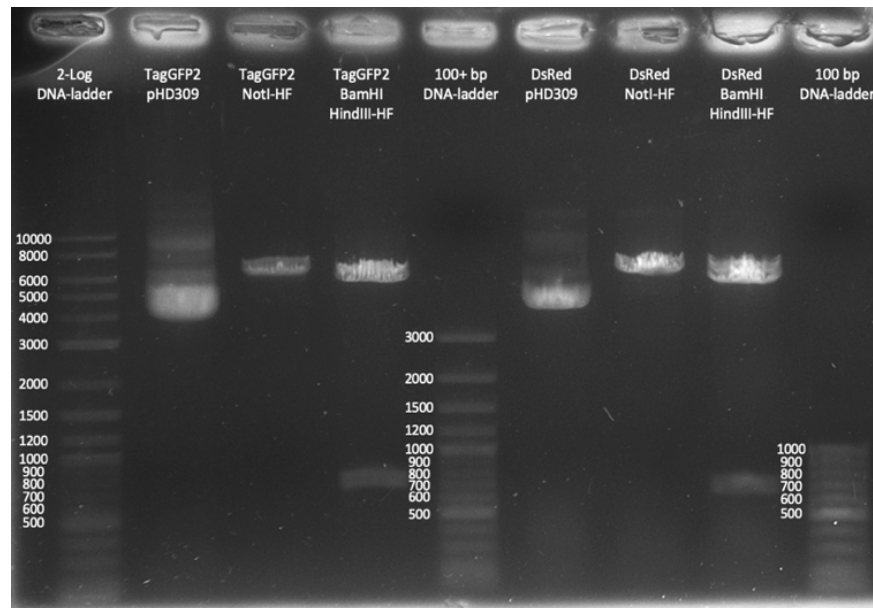


Figure 14: Agarose gel showing the bands corresponding to their DNA fragment.

The gel shows the different bands that are the result of the restriction digest with the enzymes NotI-HF, BamHI and HindIII-HF. Used DNA-ladders are 2-Log, 100+ bp and 100 bp respectively. Bands were seen at around 6500 bp, 6000 bp and 700 bp for the fragments cut with restriction enzymes.

4.4.2 Transfection with fluorescent reporter genes

The transfections using U-033 and D-023 programs did not result in the generation of any viable clones. However, the transfection carried out with the X-001 program was successful in generating a total of 16 clones. The success and efficiency of the transfection were assessed by monitoring the survival, growth, and, eventually, the expression of the fluorescent gene. The transfected trypanosomes were therefore observed via epifluorescence at different objectives. Table 3 provides an overview of the obtained clones. The clones that were visible with the smallest objective of the epifluorescence microscope, hence the most fluorescent, were then selected for further propagation. Overall, we produced clones of each strain that could be seen using the 10x or 20x objective, thus showing the stable integration of the TagGFP2 or DsRed reporter genes (Figure 15). Since we only need one LiTat 1.3 and one LiTat 1.5 strain for the ITL, preferably with a different marker, we focused on the ones that emit the most fluorescence for further use: LiTat 1.3 TagGFP2 and LiTat 1.5 DsRed.

Table 3: Clones obtained after transfection.

The table includes their source and the smallest objective that was able to visualise the trypanosomes. Since the U-033 and D-023 programs did not result in viable clones, all clones presented were obtained using the X-001 program.

Strain	Fluorescent marker	Source	Smallest objective
<i>T. b. brucei</i> LiTat 1.3	TagGFP2	<i>in vivo</i>	10x
			20x
		<i>in vitro</i>	10x (3 clones)
			20x
<i>T. b. brucei</i> LiTat 1.3	DsRed	<i>in vitro</i>	40x
			20x
			40x (4 clones)
<i>T. b. brucei</i> LiTat 1.5	TagGFP2	<i>in vivo</i>	60x
<i>T. b. brucei</i> LiTat 1.5	DsRed	<i>in vitro</i>	20x
		<i>in vivo</i>	10x
			40x

In order to compare the fluorescence intensity and visibility of the LiTat and AnTaR 1 strains, the strains were visually examined with epifluorescence. The visual comparison demonstrated that the AnTaR 1 strains were readily visible using the 10x objective, whereas a clear visual image of the LiTat strains minimally required a 20x objective. This discrepancy in visibility suggested that the LiTat strains exhibited lower fluorescence intensity, possibly attributed to the integration of fewer inserts of the TagGFP2 or DsRed genes into the β -tubulin region during transfection (Supporting figure 4).

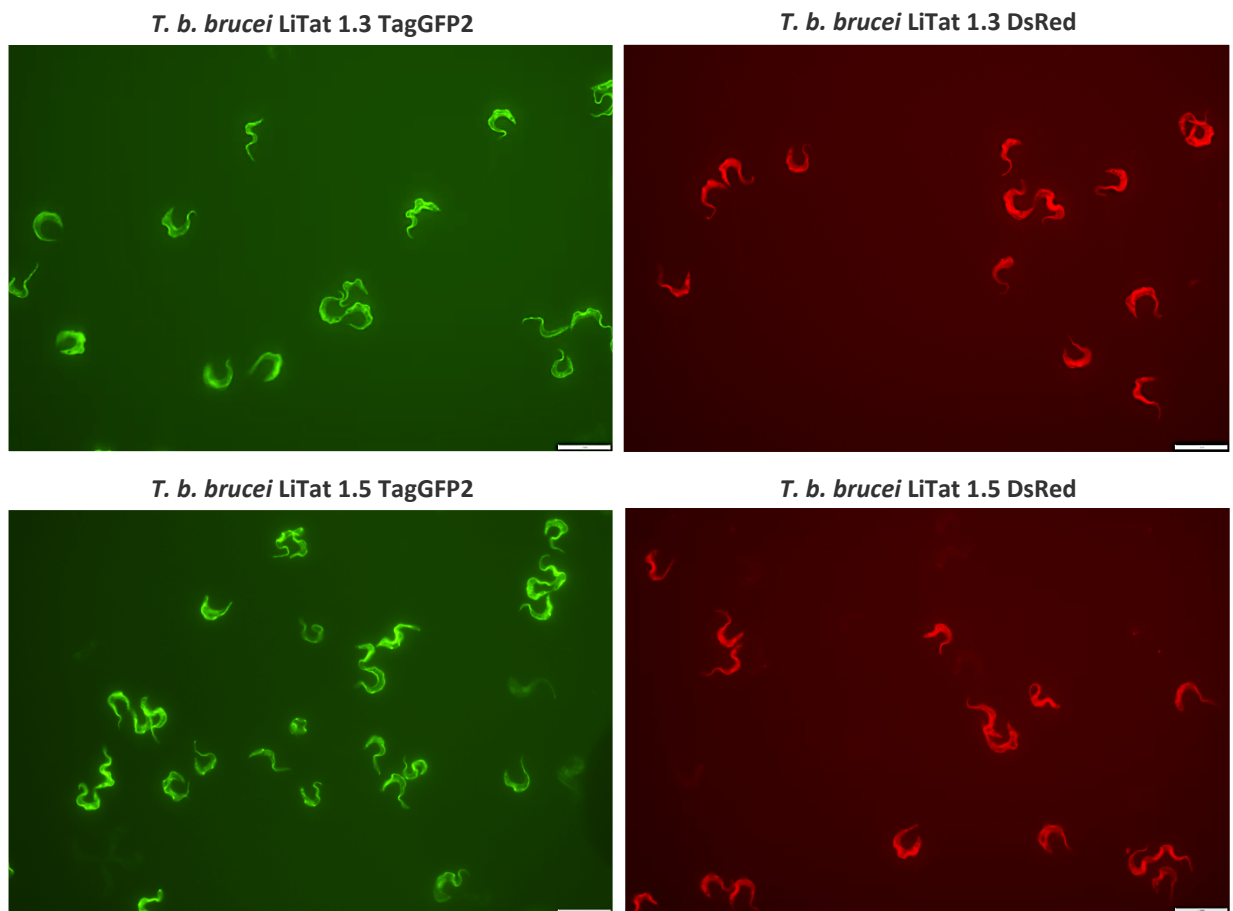


Figure 15: Fluorescent *T. b. brucei* LiTat showing the successful integration of the reporter genes (60x objective).

4.4.3 Reproducibility of the Spark experiments with the fluorescent LiTat strains

We investigated whether Spark experiments using the newly developed LiTat strains generate similar results as the ones we obtained for the fluorescent AnTaR 1 strains, the result of the fluorescent LiTat strains were compared. We observed that the instrument encountered difficulties detecting the trypanosomes, indicated by lower counts. This was particularly the case for the LiTat 1.5 DsRed strain when using the 4x objective (Figure 16). We hypothesised that this could be due to the lower fluorescence demonstrated by the LiTat strains. Since we also observed that the plateau of the linear relationship was reached earlier, we made the decision to adjust the number of trypanosomes to 4×10^5 per well.

Based on these findings, and because our ultimate goal is to make a multiplex assay, we opted to modify the Spark CYTO 400 protocol using the 10x objective, which provided higher detection capability. However, to maintain a larger field of view, we programmed the machine to count three areas within each well instead of only one (Supporting figure 6). While this adjustment allowed for enhanced detection using the 10x objective, it inevitably increased the time required per reading.

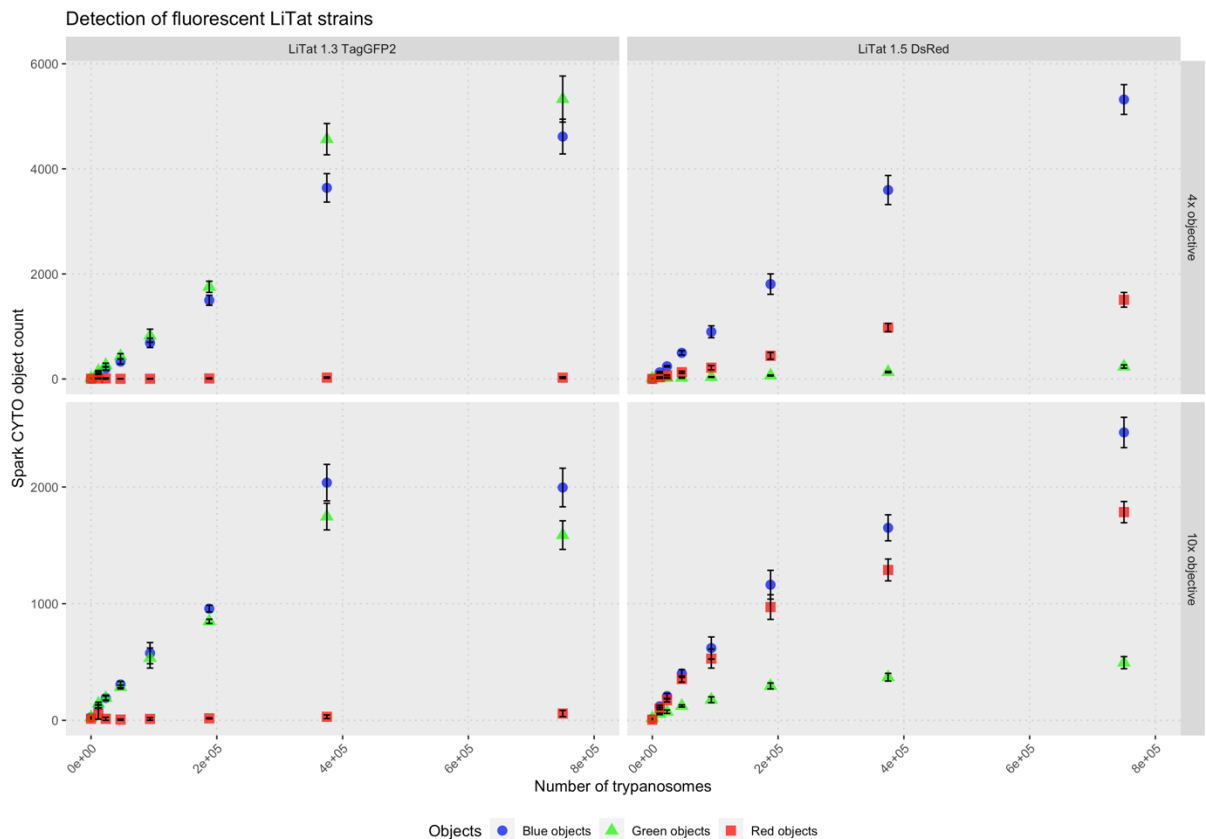


Figure 16: The Spark CYTO object count for the fluorescent LiTat strains using the 4x and 10x objectives.

The graph shows the number of objects detected by the Spark CYTO 400 when a ½ dilution of trypanosomes was placed into the reader. As demonstrated by the low counts, the instrument had difficulties detecting the LiTat 1.5 DsRed trypanosomes using the 4x objective. Error bars indicate SD.

4.5 Comparison of the fluorescent *T. b. brucei* LiTat ITL with the original ITL

Before the fluorescent trypanosomes could be used to test the high-throughput ITL, the correct VSG expression had to be confirmed again. We opted to add Hoechst (25 µg/mL) to the *in vivo* cultivated strains, so the trypanolytic activity could also be visualised using epifluorescence microscopy. Furthermore, we performed an ITL on a mixture of both strains (same concentration and volume) to get a first impression of how a multiplex assay would work. The ITL performed on the LiTat 1.3 TagGFP2 and LiTat 1.5 DsRed strains confirmed their anticipated response, as indicated by a noticeable decrease in trypanosomes within the wells containing the corresponding specific antisera (Table 4). Additionally, the epifluorescence reading allowed for the visualisation of the trypanolytic action without being hindered by erythrocytes or other extraneous particles, demonstrating the potential effectiveness of the respective strains in a fluorescent ITL. In the multiplex assay using a mixture of both strains, the same pattern of lysis was observed. The assessment of lysis was facilitated by observing the number of Hoechst-stained trypanosomes first. Upon examining these Hoechst-stained cells and subsequently switching to the green or red channel of the epifluorescence microscope, it was possible to differentiate between TagGFP2 and DsRed labelled trypanosomes. If the blue-stained cells (i.e. nucleus and kinetoplast) exclusively displayed red fluorescence, it indicated the occurrence of LiTat 1.3 strain-specific lysis, and conversely, if they exhibited green fluorescence, LiTat 1.5 strain-specific lysis was observed. When both strains were lysed, only their blue-stained nuclei could be observed, with complete disappearance of the other fluorescent labels. In cases where both green and red trypanosomes were observed, it indicated the absence of lysis (Figure 17).

These results confirm that integrating fluorescent reporter genes in the transgenic trypanosome strains did not hamper the correct VSG expression and demonstrate the feasibility of utilising these strains in a high-throughput ITL.

Table 4: ITL results of the fluorescent *T. b. brucei* LiTat strains.

The results show the percentage of trypanolysis (i.e., 100% = all trypanosomes were lysed or dead, 0% = all trypanosomes were alive after 90 min of incubation.). The results prove that the transfection did not jeopardise the correct VSG expression of the strains.

<i>T. b. brucei</i> LiTat 1.3 TagGFP2			<i>T. b. brucei</i> LiTat 1.5 DsRed		
GPS	PSG	α-VSG 117	GPS	PSG	α-VSG 117
0	0	0	0	0	0
α-VSG 221	NRS	R-008 - AnTat1.1/6	α-VSG 221	NRS	R-008 - AnTat1.1/6
0	0	0	0	0	0
R008 - AnTat1.1/26	NHS - ND	NHS - NVR	R008 - AnTat1.1/26	NHS - ND	NHS - NVR
100	0	0	80	0	0
NHS - PB	LiTat 1.1 - K446/7	LiTat 1.2 - K478/6	NHS - PB	LiTat 1.1 - K446/7	LiTat 1.2 - K478/6
0	0	0	0	0	0
LiTat 1.3 - K551/9	LiTat 1.3 - K551/12	LiTat 1.3 - K552/9	LiTat 1.3 - K551/9	LiTat 1.3 - K551/12	LiTat 1.3 - K552/9
100	100	100	0	100	0
LiTat 1.3 - K552/12	LiTat 1.3 - K685/28	LiTat 1.4 - K372/6	LiTat 1.3 - K552/12	LiTat 1.3 - K685/28	LiTat 1.4 - K372/6
100	100	0	100	100	0
LiTat 1.5 - K377/6	LiTat 1.5 - K378/14	LiTat 1.5 - K378/120	LiTat 1.5 - K377/6	LiTat 1.5 - K378/14	LiTat 1.5 - K378/120
0	100	100	100	100	100
LiTat 1.6 - K425/4	LiTat 1.6 - K425/5	LiTat 1.6 - K425/8	LiTat 1.6 - K425/4	LiTat 1.6 - K425/5	LiTat 1.6 - K425/8
0	0	0	0	0	0
LiTat 1.6 - K425/20	LiTat 1.7 - K463/6	LiTat 1.8 - K465/6	LiTat 1.6 - K425/20	LiTat 1.7 - K463/6	LiTat 1.8 - K465/6
100	0	0	100	0	0
LiTat 1.9 - K466/6	LiTat 1.10 - K604/6	/	LiTat 1.9 - K466/6	LiTat 1.10 - K604/6	/
0	20	/	0	0	/

To evaluate the high throughput ITL using the Spark CYTO 400, we performed the rabbit sera assay with the fluorescent LiTat 1.3 and 1.5 strains. The experimental setup followed a protocol similar to the conventional ITL, ensuring consistency in the comparison. Each well's initial trypanosome starting count was used as a reference point for calculating the normalised number of living trypanosomes. A positive test result was determined if, after 90 minutes, the relative trypanosome count was less than 50% of the starting point. This is similar to the original ITL, although each well was used as its own reference here. To analyse the results, a graphical representation was made (Figure 18; Figure 19), which allowed for clear visualisation of the kinetics of the lytic activity, enabling a reliable determination of the test outcomes based on the predefined threshold. Overall, all

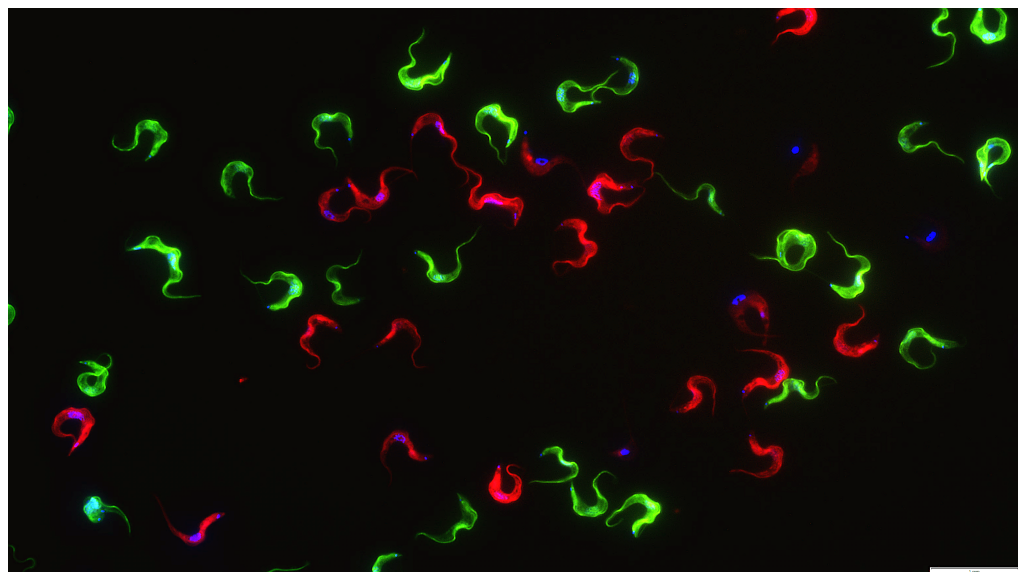


Figure 17: Epifluorescence image of LiTat 1.3 TagGFP2 and LiTat 1.5 DsRed stained with Hoechst (60x objective).

obtained results aligned with the results of the conventional ITL. The graphs of the wells with individual LiTat strains (top two rows) show the same reaction and kinetics as the combination of both strains. However, an unexpected outcome was observed for LiTat 1.7: we noticed a significant drop in trypanosome counts to zero. We suspect that this is more likely attributed to a potential failure of the machine in detecting the trypanosomes rather than an actual lytic action of the serum. Additionally, we observed a lytic effect of the LiTat 1.10 serum. Although the underlying mechanism remains unknown, it is worth noting that this serum already demonstrated some lytic activity for the *T. b. brucei* LiTat strains in the earlier performed trypanolysis assays.

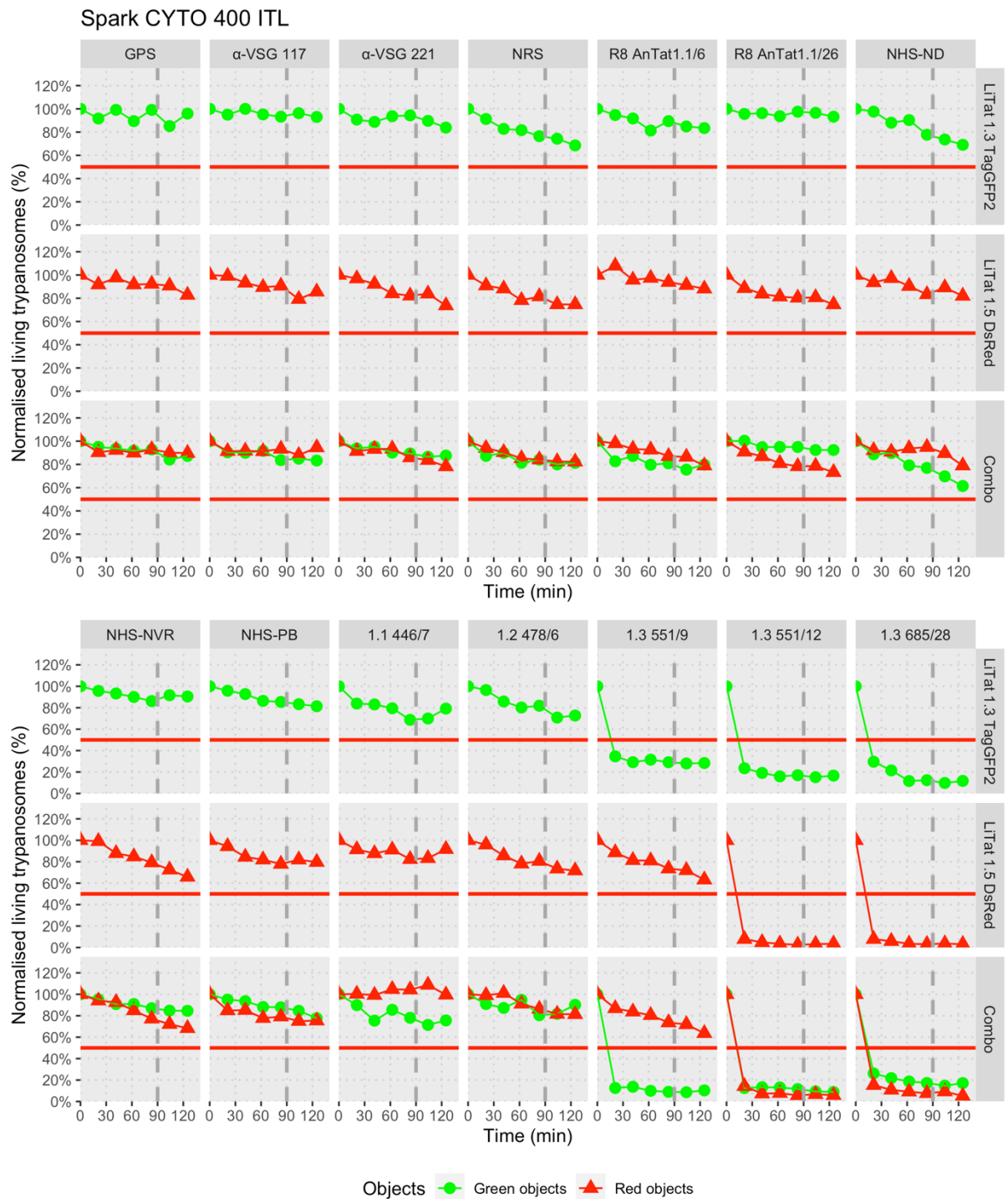


Figure 18: Kinetic ITL of the fluorescent LiTat strains against the panel of NHS and rabbit sera (1).

The *T. b. brucei* LiTat 1.3 TagGFP2 and LiTat 1.5 DsRed strains were subjected to an ITL individually and in a combination. Trypanosome counts were normalised based on their initial measurement. The dashed line shows the endpoint of the normal ITL (i.e., 90 min), while the red line shows the threshold of 50% lysis.

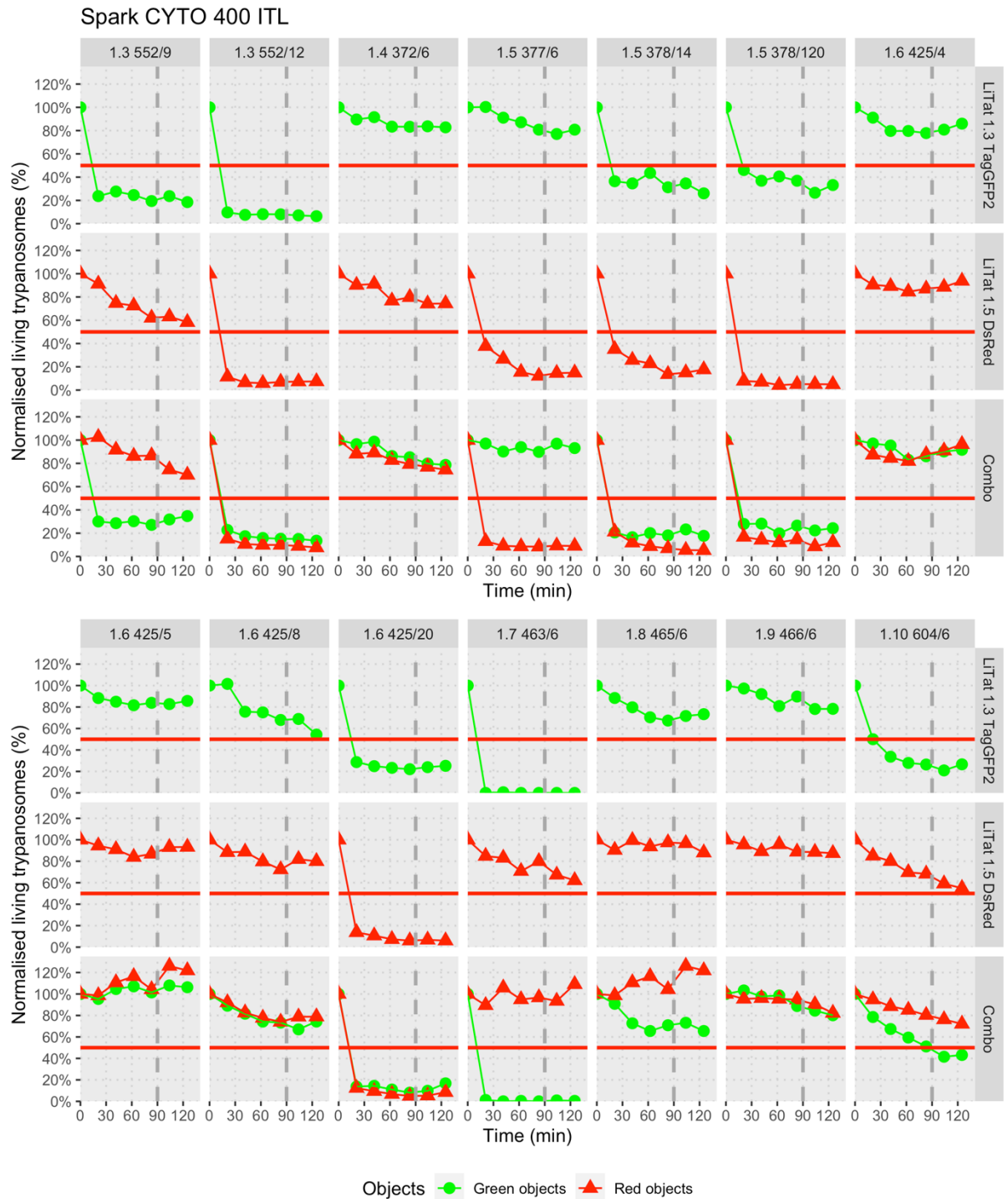


Figure 19: Kinetic ITL of the fluorescent *LiTat* strains against the panel of NHS and rabbit sera (2).

The *T. b. brucei* *LiTat* 1.3 TagGFP2 and *LiTat* 1.5 DsRed strains were subjected to an ITL individually and in a combination. Trypanosomes counts were normalised based on their initial measurement. The dashed line shows the endpoint of the normal ITL (i.e., 90 min), while the red line shows the threshold of 50% lysis.

4.6 Validation of the fluorescent *T. b. brucei* ITL using human samples

To validate the fluorescent strains, a subset of human samples from the HAT-PolyB diagnostic study was selected to assess the feasibility and effectiveness of a multiplex ITL with automated readout. To demonstrate the method's capability, we specifically included cases that tested positive for only one *LiTat* variant. The validation experiment was conducted twice (on different days), and the obtained results were compared to those of a previously performed ITL.

Figure 20 gives an example of how the serum of a positive patient can cause the fluorescent trypanosomes to almost completely disappear after 90 min in comparison to a healthy control. The results demonstrated substantial similarity to the conventional ITL results. A representative subset of the results is depicted in Figure 21, while the complete set of results can be found in Supporting figure 8-10. Moreover, the findings showed consistent trends and outcomes between the two experimental runs, indicating the reliability and reproducibility of the multiplex ITL. Interestingly, certain samples of run 2 exhibited delayed positivity compared to the first run. This difference could be attributed to using different batches of GPS, potentially with different concentrations of complement.

In cases where deviations from the conventional ITL were observed, it was often due to significantly lower trypanosome counts in the wells. On average, each well started with a count exceeding 2000 trypanosomes. However, the starting count in some wells was only a few hundred trypanosomes. This inconsistency could be attributed to factors present in the serum, potentially causing fluorescence signal blocking and reduced detection. Additionally, some wells in the first run did not yield results due to autofocus errors during the reading process. While the exact cause of these errors remains unknown, a potential reason could be that particles such as dust adhered to the clear bottom of the plate.

Cohen's kappa coefficient was calculated to evaluate the level of agreement between the two methods (Table 5). The analysis revealed a significant level of agreement, with most kappa values exceeding 0.8, indicating an 'almost perfect' agreement between the multiplex method and the previously conducted ITL. Moreover, when comparing the kappa values from the combination of both LiTat 1.3 and 1.5 strains to those from each strain separately, the combination resulted in higher kappa values. This indicates that, even though some results were contradictory, the combination resulted in an improved agreement with the conventional ITL. Finally, as shown already by the graphical analysis, the kappa values also indicate that the trypanolysis might require more time than the standard 90 min. The degrees of agreement increased over time, reaching their peak at 110 min, suggesting that the optimal readout time should be adjusted to approximately 110 min after incubation.

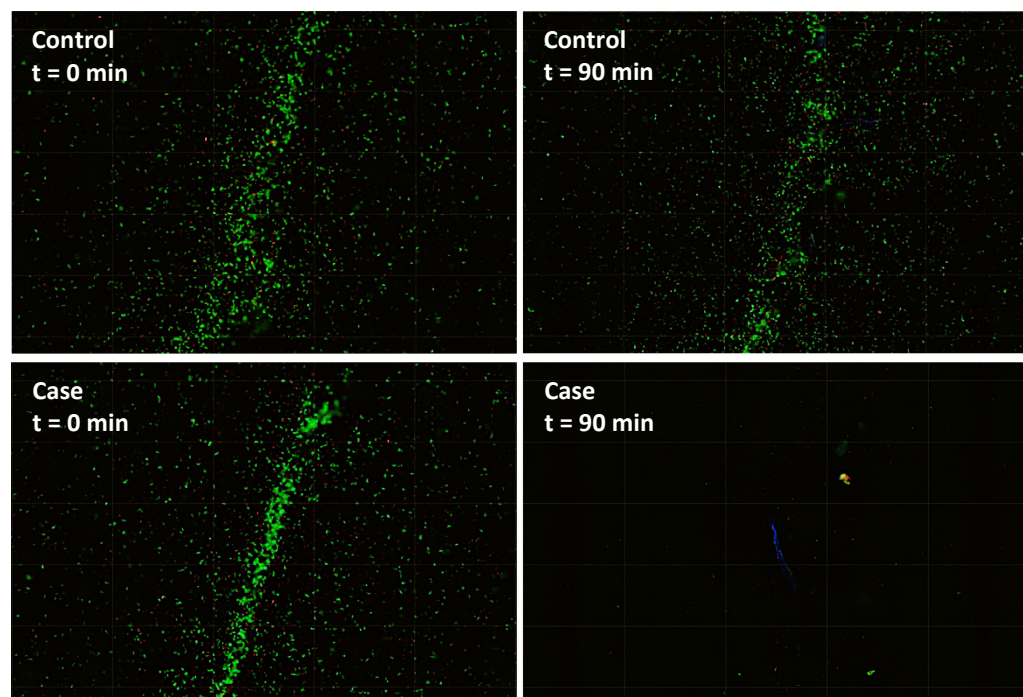


Figure 20: Spark CYTO 400 overlay images of the HAT-PolyB trypanolysis

The images show the trypanolytic effect on the fluorescent LiTat trypanosomes at the start of the test ($t = 0$ min) and at the 90-min endpoint ($t = 90$ min). The number of trypanosomes clearly drops after incubation in serum of a positive patient. The images were obtained via the Spark's ImageAnalyzer software.

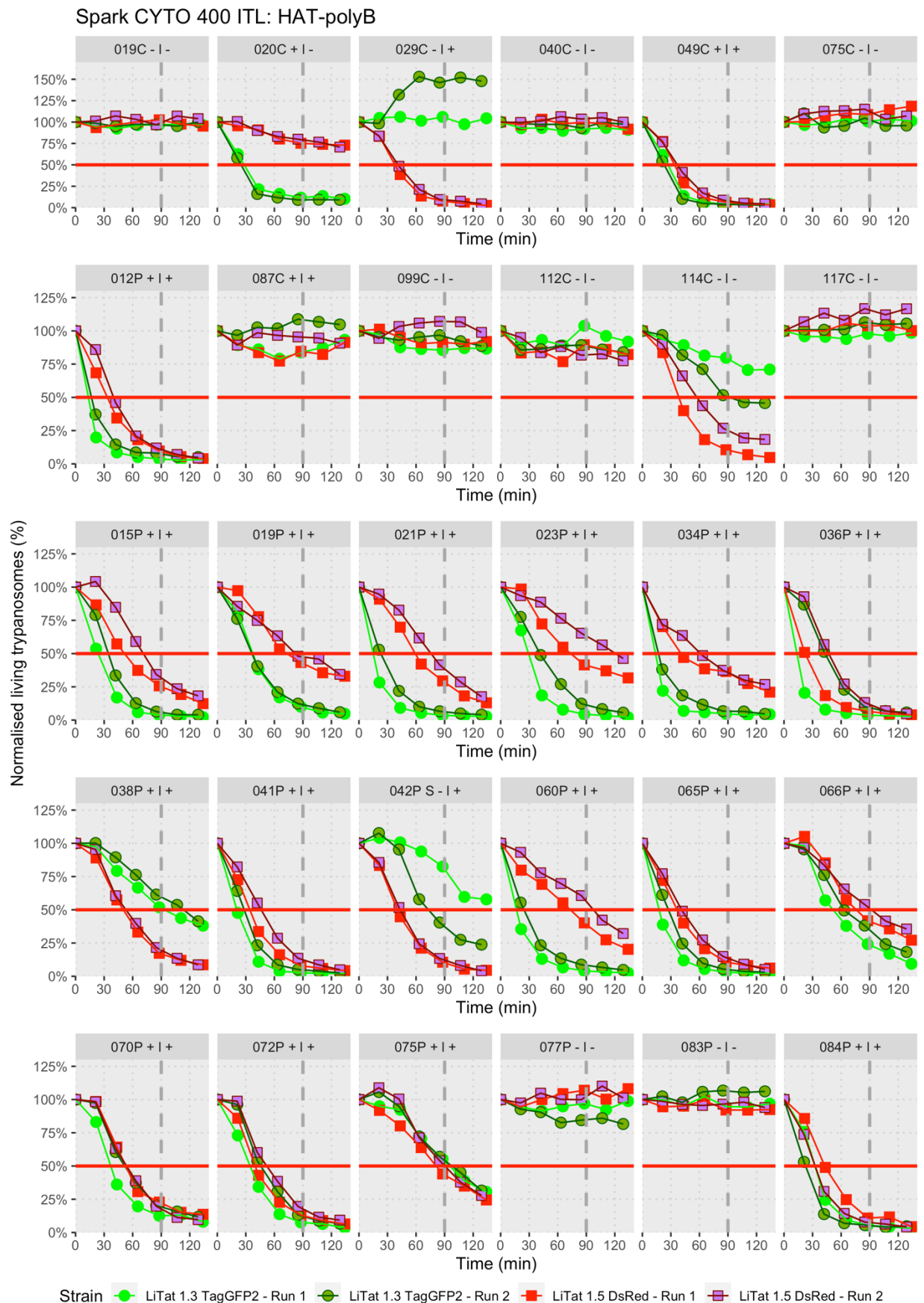


Figure 21: ITL results of HAT-PolyB samples.

A representative subset of the kinetic ITL results. The symbols behind the sample ID indicate the original result of the ITL: “- I -” means negative for LiTat 1.3 and 1.5, “+ I -” means positive for LiTat 1.3 and negative for 1.5, etc. Trypanosomes counts were normalised based on their initial measurement. The dashed line shows the endpoint of the normal ITL (i.e., 90 min), while the red line shows the threshold of 50% lysis.

Table 5: Cohen's kappa for the fluorescent ITL with the Spark CYTO 400.

Cohen's kappa values were calculated with the results of the conventional ITL as reference. P_o represents the proportion of observed agreement, and P_c is the proportion of agreements expected by chance.

Strain	Run	Time (min)	P_o	P_c	Cohen's kappa
LiTat 1.3 TagGFP2	Run 1	90	0,93	0,51	0,85
		110	0,95	0,50	0,90
		130	0,94	0,50	0,88
	Run 2	90	0,90	0,51	0,79
		110	0,91	0,51	0,81
		130	0,92	0,50	0,84
LiTat 1.5 DsRed	Run 1	90	0,95	0,50	0,96
		110	0,95	0,50	0,97
		130	0,95	0,50	0,91
	Run 2	90	0,83	0,51	0,66
		110	0,92	0,50	0,84
		130	0,94	0,50	0,89
Combination	Run 1	90	0,96	0,50	0,93
		110	0,96	0,50	0,93
		130	0,96	0,50	0,93
	Run 2	90	0,94	0,50	0,88
		110	0,98	0,50	0,95
		130	0,97	0,50	0,93

5. Discussion

Human African trypanosomiasis, a life-threatening parasitic infection, is predominantly caused by *Trypanosoma brucei gambiense*. Despite being responsible for devastating epidemics throughout the 20th century, efforts in disease control and treatments made it a realistic target to eliminate the transmission of the disease by 2030. However, even when this milestone is reached, endemic countries must maintain dedicated surveillance systems to prevent potential re-emergence or re-introduction of the disease. Therefore, sensitive, and highly specific diagnosis remains crucial. To date, the current serodiagnostic tests rely on cultivating highly virulent parasites, posing significant biohazard safety risks, and demanding a high number of animals. Moreover, the most specific serological test, the ITL, has limitations regarding low throughput and reliance on microscopy for parasite lysis confirmation [1, 74]. In this thesis, we aimed to address these drawbacks by making an ITL based on *in vitro* cultivated fluorescent transgenic *Trypanosoma brucei brucei* strains expressing LiTat 1.3 or LiTat 1.5 that could be detected by a high-throughput cell imaging system such as the Spark CYTO 400.

In order to validate the VSG expression of our transgenic strains, we utilised a panel of rabbit sera collected at various time points post-infection. As shown by previous sub-inoculation experiments in rabbits by Gray, the antigenic character of a strain remains consistent up to the ninth day of infection, as evidenced by the ability of the sera to agglutinate substrains isolated between days three and seven. However, after the ninth day, serologically distinct substrains were isolated. This suggests that the ninth day may serve as a critical turning point, beyond which sera no longer exhibit monovalent reactivity. In other words, rabbit sera demonstrate monovalent reactivity until the ninth day of infection, transitioning to polyvalent reactivity thereafter [75]. Based on these findings, our transgenic strains were only lysed by sera that could be considered as monovalent (specifically directed against the corresponding strain) or polyvalent. Moreover, we showed that the *T. b. brucei* LiTat 1.3 and LiTat 1.5 exhibited a reliable response to specific immuno-trypanolysis comparable to the *T. b. gambiense* strains. This finding suggests that these transgenic strains have the potential to serve as safe and suitable replacements for the currently used *gambiense* strains in the ITL. Using *T. b. brucei* strains offers several advantages over the conventional approach. Persons working with *T. b. gambiense* are at potential risk of getting infected when working with these highly virulent parasites. These infections can occur through various routes of exposure, highlighting the importance of implementing rigorous safety measures when handling these parasites. Since trypanosomes multiply in the host's bloodstream, even a small inoculum is enough to get infected. The highest risk for accidental infections is when laboratory personnel inoculate mice or rats to cultivate specific

strains. Moreover, microscopic observations of the parasites (e.g., to check the lysis in the ITL) also impose a potential risk for accidental infection when staff cut their hands with the glass coverslip [76]. Implementing *in vitro* cultures and using the transgenic non-human pathogenic *T. b. brucei* strain would minimise the risk of infection by prick accidents. Additionally, the *in vitro* cultivation of these strains would eliminate the need for maintaining an animal facility, reducing the associated costs and ethical concerns. This would lead to less laboratory animal use, thereby complying with the 3Rs (replacement, reduction, and refinement) and allows to perform the ITL in more resource-limited settings. Apart from general laboratory equipment (e.g., glassware to prepare culture media, culture flasks, etc.), the only things needed are an incubator to keep the cultures at 37°C and 5% CO₂ and a microbiological safety cabinet class II to avoid contamination of cultures. However, further studies are still needed to investigate the scalability and cost-effectiveness of implementing these transgenic strains and *in vitro* cultures in routine diagnostic settings.

In light of enabling parallel testing of numerous samples, the adapted ITL involved integrating fluorescent reporter genes in the transgenic strains. As previously shown, the pH309 plasmid allowed the effective integration of TagGFP2 and DsRed genes into the β -tubulin region [64]. By reading the test through epifluorescence, we could observe the strain-specific lysis for the LiTat strains in a multiplex format, highlighting the potential benefits of incorporating fluorescent trypanosomes in the diagnostic workflow. Observing the number of Hoechst-stained trypanosomes and, afterwards, differentiating between TagGFP2 and DsRed-labelled trypanosomes showed a convenient way to check both strains simultaneously. This basic approach for a multiplex ITL only requires an epifluorescence microscope, thus could be a possibility in laboratories with fewer resources. An additional advantage is that fluorescence-based observation encounters less hinder from particles such as erythrocytes or artefacts in low-quality slides that could be mistakenly interpreted as lysed trypanosomes.

In addition, the experiment utilising the Spark CYTO 400 demonstrated the ability to analyse multiple samples at the same time, providing a rapid and efficient evaluation of the immune trypanolysis process. The results revealed a high degree of concordance between the adapted ITL and the conventional ITL, as demonstrated by kappa values above 0.8, indicating strong agreement. Additionally, a remarkable consistency was observed between the two experimental runs conducted using the HAT-PolyB samples. The results consistently aligned across the majority of the samples, reinforcing the reproducibility of the outcomes. Specifically, when a sample was determined to be negative in the first run, the corresponding analysis in the second run yielded the same negative result. Similarly, in cases where samples tested positive, both runs consistently confirmed the presence of trypanolysis. Furthermore, the kinetics of the lytic activity exhibited similarity between the two experimental runs. When the lytic action was rapid in run 1, the same response was observed in run 2. Likewise, when the lytic action appeared to be slower in run 1, a corresponding slower kinetics pattern was observed in run 2. However, some samples in the second run showed delayed positivity compared to the first run, possibly due to different GPS batches. Distinct batches may contain varying complement levels, impacting the efficiency of complement-mediated lysis [77]. This delay in reaching the 50% threshold and the higher Cohen's kappa values suggest extending the reading window beyond the designated 90 min. Even in the conventional ITL, not all samples are read out at exactly 90 min. Due to the time required for microscopic evaluation, a considerable amount of time passes before the ITL-reader reaches the final samples on the 96-well plate. This delay can result in samples that may be scored as negative after 90 min due to the relatively slower kinetics of these specific samples. However, the implementation of the new ITL utilising the cell imager provides a significant advantage by offering improved standardisation of the reading time. This minimises significant differences in the timing of the readings, enhancing the reliability and consistency of the test results.

Overall, the new ITL method showed advantages such as reduced turnaround time, improved efficiency, and scalability, making it suitable for high-throughput testing. The number of trypanosomes needed per automated test was also found to be lower than the conventional ITL, preventing problems concerning lower parasite concentrations known to be associated with *in vitro* cultures. Moreover, this method reduces subjectivity and potential resuspension errors compared to the conventional ITL. In the conventional ITL, 8 μ L of each well is placed on a microscopic slide, and then the presence or absence of lysis is checked by observing ten microscopic fields. However, when wells are not adequately resuspended, the risk exists that unreliable results are reported.

Apart from this, the reading is still performed by estimating the percentage of lysis. When all parasites are lysed, this is quite straightforward. However, when some parasites are still alive, the ITL is prone to subjectivity. This subjective nature arises from different individuals having different levels of expertise and experience in identifying and quantifying the degree of lysis. Factors such as personal judgment, visual precision, and fatigue can influence the accuracy of the results. Using machine-based detectors such as the Spark CYTO 400 could address these drawbacks. An automatic result interpretation also enables data analysis at the laboratory technician's convenience, eliminating the constraint of being bound to the 90-min time frame. This enhances flexibility, allowing the technician to engage in other projects simultaneously.

Despite its advantages, the use of the Spark instrument has certain shortcomings. One limitation is the reliance on the device itself, as we observed during our study. Occasionally, the device encountered errors or failed to accurately count the trypanosomes, resulting in lower counts. Factors in the serum may block the fluorescent signal, or observing only three fields per well might give unreliable results. However, the Spark system still allows visually examining any wells that gave doubtful results using microscopy (following the conventional ITL) or with the Live Viewer option. A limitation of our study is the need for further validation on a larger human serum sample size. Due to time constraints, we could not test additional samples extensively. While our results are promising, it is essential to validate them on a larger scale to ensure the reliability and reproducibility of the findings. Moreover, we relied on reported ITL results from the HAT-PolyB study as a reference instead of conducting the ITL again. Due to sample degradation over time or possible false reading results during the initial ITL, potential discordance might be explained. Conducting parallel testing on fresh samples would provide more robust and reliable data. Additionally, our study did not thoroughly test the scalability of the *in vitro* cultures. The current methodology requires scaling up the cultures on a weekly basis. However, we encountered challenges in predicting and ensuring consistent growth of the cultures. This unpredictability led to delays in experiments, as the cultures were not always sufficiently grown. Exploring and optimising the scalability would be essential for implementing the alternative diagnostic method. Finally, the cost-effectiveness was not explicitly considered in our study. For any alternative method to be a viable replacement for the conventional ITL, it should not impose excessive financial burdens. A clear view of the cost of *in vitro* culture and the purchase and materials needed for using the Spark CYTO cell imager would be necessary to assess the economic feasibility of implementing the alternative diagnostic approach on a larger scale.

5.1.1 The trypanolytic activity by ApoL1 does not interfere with the *brucei*-ITL protocol

Since the fluorescent *T. b. brucei* strains are exposed to the trypanolytic activity of ApoL1 present in NHS, we initially expected that we had to encounter this lytic activity in a way to stall or inhibit its action. However, our ITL and Spark experiments findings revealed that ApoL1 did not exhibit any significant impact on the *T. b. brucei* strains under the experimental conditions employed. During the experiments, the cells were monitored after or for at least 90 minutes. Despite the presence of ApoL1 in NHS, no noticeable decline in viability was observed. Our results are supported by Vanhollebeke *et al.*, who showed that, at the concentration of NHS used in the ITL (i.e., 25%), the percentage of viable trypanosomes remains stable for approximately two hours [78]. These findings indicate that TLFs do not induce a significant trypanolytic activity that would impede using the transgenic *T. b. brucei* strains in the ITL. However, if subsequent testing of multiple samples contradicts this hypothesis, the following alternative strategies could be explored to inhibit the activity of ApoL1: Lowering the temperature of the trypanosome suspension is known to reduce the activity of ApoL1. At 4°C, TLFs can bind to the flagellar pocket, but endocytosis is inhibited. Furthermore, studies showed that lysis does not occur at temperatures below 17°C regardless of the incubation time or concentration of TLFs because the TLF stays in the small coated vesicles [79]. Another possibility is to add the basifying agents ammonium chloride or chloroquine to stall or inhibit the trypanolytic activity. Hence, raising the endolysosomal pH can block acidification and cell lysis [79-81].

5.1.2 Positioning within the WHO's Target Product Profile 4

As efforts are made to achieve HAT elimination goals, there remains a crucial need for large-scale testing of populations at risk to verify the absence of *T. b. gambiense* transmission. Since the current diagnostic tools are

often complex and resource-demanding, the WHO has formulated Target Product Profile 4 (TPP4): a high-throughput test for verification of elimination. The underlying concept behind this TPP4 is developing a new diagnostic method that would allow samples to be collected by non-specialised personnel in remote rural areas and subsequently shipped to reference laboratories for high-throughput analysis.

One of the key considerations for a suitable TPP4 diagnostic method is its applicability in resource-limited settings and local sample analysis. By circumventing the reliance on *in vivo* cultivation, our test can significantly reduce the resource requirements and complexity associated with managing an animal facility. Moreover, the *in vitro* cultivation of the *T. b. brucei* strains provides a safer alternative, needing less specialised personnel. However, the affordability of the Spark CYTO 400 device may limit its accessibility to all laboratories. Therefore, additional research is required to explore cost-effective instruments that could substitute for the Spark. In terms of throughput, the multiplexing of the readout ensures that the time needed per sample can be considerably decreased. Although the data processing after the Spark reading now requires a considerable amount of time, the time per analysis can still be reduced once a standardised protocol is developed. This scalability is crucial for the high throughput requirements of the TPP4 diagnostic method, where a large number of samples need to be processed efficiently. With regard to the assay's performance, the technique should demonstrate a sensitivity equal to or greater than the most sensitive parasitological tests currently in use. Furthermore, the method should exhibit high repeatability and reproducibility, as indicated by a kappa value above 0.8 and preferably above 0.9. As shown by our calculation of Cohen's kappa for our subset of HAT-PolyB samples, the fluorescent ITL could meet these requirements [74].

In summary, our adapted multiplex ITL addresses several critical elements of the TPP4 outlined by the WHO. As such, the test might emerge as a promising alternative next to iELISA for TPP4. Although further optimisation is still necessary, this research might contribute to fulfilling the target product profile requirements for an effective and accessible diagnostic method for verification of elimination programs.

5.1.3 Future prospects

Several potential paths for further investigation and improvements can be considered. One area of interest lies in purifying the recombinant *gambiense* VSGs LiTat 1.3 and LiTat 1.5 expressed by the transgenic *T. b. brucei* strains. Since some serological tests such as RDTs and ELISA rely on the cultivation of virulent live trypanosomes in rodents for VSG production, these *in vitro* cultivated non-human infective strains could be a worthy replacement. The entire verification process would involve the purification using fast protein liquid chromatography (FPLC), sodium dodecyl sulphate polyacrylamide gel electrophoresis (SDS-PAGE) and Western blotting, indirect ELISA and finally, validation using human sera to know whether these rVSGs could be used.

Furthermore, developing a fluorescent multiplex ITL allows for additional research and enhancements. First, this new method should be tested and improved, preferentially using more human samples. Later, including other strains such as *T. b. brucei* expressing LiTat 1.6, fluorescently labelled with a different marker (e.g., TagBFP2), could allow the parallel detection of infected patients that do not (adequately) react to LiTat 1.3 and 1.5. This approach has already been implemented in the LATEX/*T. b. gambiense* and ELISA/*T. b. gambiense* assays, and could possibly help to increase sensitivity [47, 82, 83]. Moreover, LiTat 1.6 was found to be an important VAT for the serological testing of pigs. In an effort to eliminate HAT, the ITL might be a promising tool to investigate the involvement of animal reservoirs. Ilboudo *et al.* revealed that antibodies against LiTat 1.6 were the first to be detected in pigs after two weeks of infection and that these antibodies persisted consistently over time. Contrarily, anti-LiTat 1.3 and LiTat 1.5 antibodies fluctuated over time. This differs from human infections, where trypanolysis remains positive throughout the infection and even after treatment. The study emphasises that LiTat 1.3 and LiTat 1.5 trypanolysis are not specific in pigs and cannot be relied upon as definitive tools for identifying potential animal reservoirs of *T. b. gambiense*. However, the LiTat 1.6 ITL can be used as serological test, but without being able to differentiate between *T. b. gambiense* and *T. b. brucei* [84]. Furthermore, considering the inclusion of a strain expressing a VSG that does not elicit any reaction in patients could allow to distinguish between potentially lytic factors (e.g., in the patient's serum or GPS) that are not related to an acquired infection, avoiding false positives. Although further investigation is needed to determine which VSG could fulfil this

criterion, Rode Trypanozoon antigen type (RoTat, *T. evansi*) 1.2 (surra) could be a potential candidate. In addition, considering the possibility of developing this test for trypanosome species that are pathogenic to animals could help in the diagnosis of animal trypanosomiasis.

Alternative methods, such as flow cytometry in 96-well format or other cell imaging readers like Cytation C10 (BioTek) or Celloger Mini Plus (Curiosis), might offer other advantages over the Spark CYTO 400 in terms of throughput, acquisition cost, increased flexibility in fluorescent labels, data analysis, and potential for automation. Therefore, exploring and comparing alternatives that may offer even greater specificity or enable high-throughput testing in more resource-limited locations would be worthwhile.

Another direction worth exploring is finding a way to work with dried blood spots (DBS). Currently, the test was only performed on serum, but DBS offer advantages in terms of simplicity, shipping, and long-term stability of samples. Integrating DBS into the workflow would enable sample collection in remote areas without the need for sophisticated storage and transportation logistics, thus fulfilling more to TPP4 [45]. However, the challenge is that the filter paper discs would need to be removed from the wells, using only the eluate for the reading with the Spark CYTO 400. Since the samples are excited via the bottom of the plate, the DBS can hinder the excitation of the trypanosomes or autofluorescence from the paper might interfere with the fluorescent signal generated by the sample. Therefore, this method would require extraction from the filter papers, though it is not always certain whether all the necessary antibodies effectively dissociate from the DBS.

6. Conclusion

In summary, this research contributes to the development of a safer, animal-friendly, and high-throughput diagnostic method for HAT, aligning with the ongoing efforts to eliminate the transmission of the disease. The successful validation of the transgenic *T. b. brucei* strains opens up new possibilities for safer serodiagnosis of HAT. By substituting animal cultivation with *in vitro* culture, HAT diagnosis can be performed more ethically. Moreover, the requirement for less equipment enables its implementation in a wider range of locations, particularly in resource-limited settings where access to animal facilities might be difficult. The automated kinetic ITL combining fluorescent LiTat 1.3 and LiTat 1.5 trypanosomes showed promising results that align with the conventional ITL. However, further investigation and protocol improvements are required to explore the kinetics of the lytic reaction, the potential benefits of including other VATs, and address challenges related to the use of DBS, device errors and fluorescence signal blocking. With ongoing research and protocol refinements, this method can potentially enhance the efficiency of HAT diagnostics on the road to eliminating the disease.

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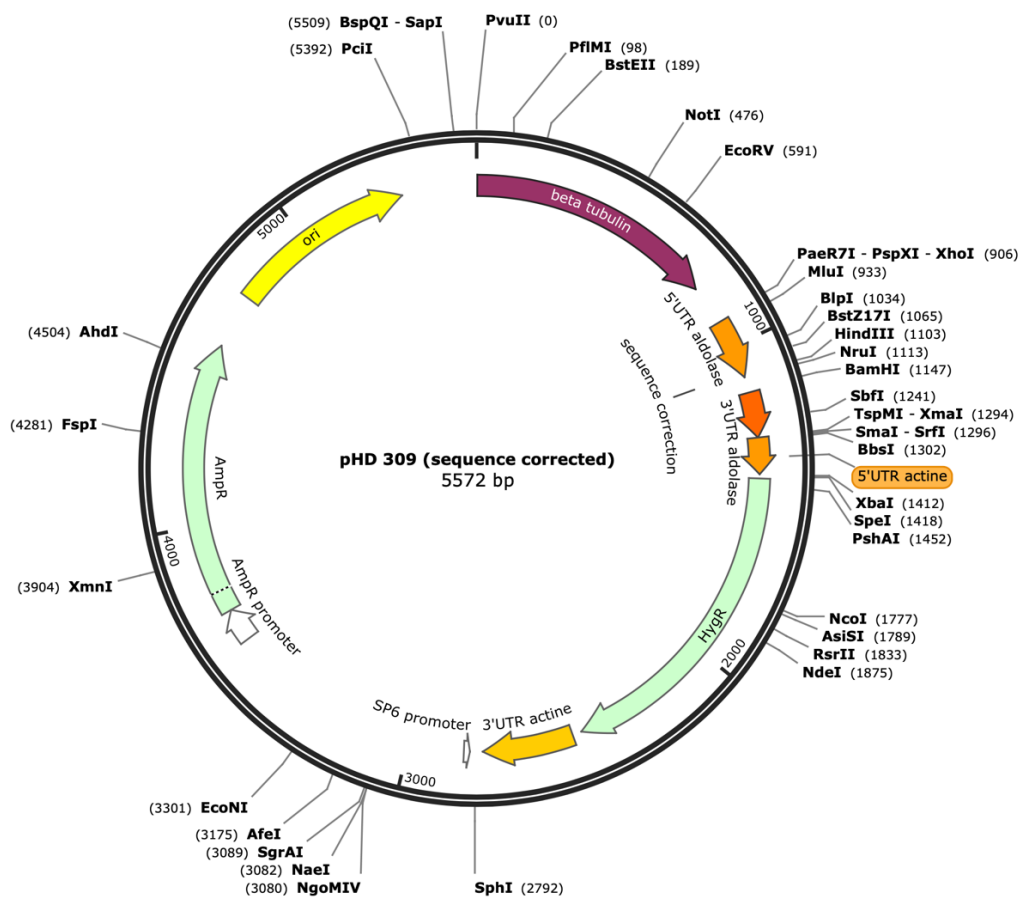
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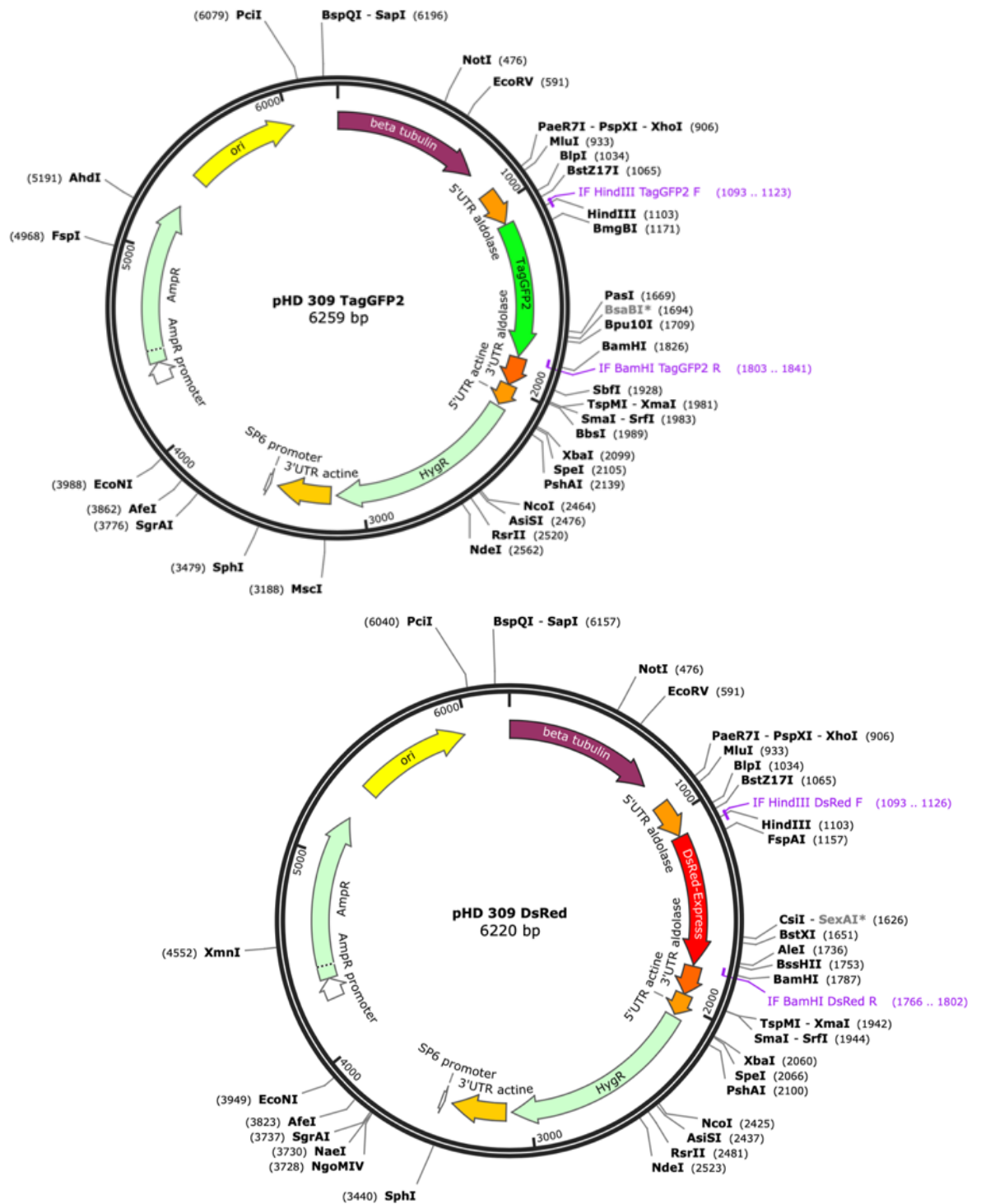
Addendum

Phylum	Protozoa						
Order	Kinetoplastida						
Suborder	Trypanosomatina						
Family	Trypanosomatidae						
Genus	Trypanosoma						
Subgenus	Herpetosoma	Mega-trypanum	Schizo-trypanum	Duttonella	Nannomonas	Trypanozoon	Pycnomonas
Species	<i>T. theleri</i>	<i>T. rangeli</i> <i>T. lewisi</i> <i>T. musculi</i>	<i>T. cruzi</i>	<i>T. vivax</i> <i>T. uniforme</i>	<i>T. congolense</i> <i>T. simiae</i> <i>T. godfreyi</i>	<i>T. equiperdum</i> <i>T. evansi</i> <i>T. brucei</i>	<i>T. suis</i>
Subspecies		Stercoraria		Salivaria		<i>T.b. brucei</i> <i>T.b. rhodesiense</i> <i>T.b. gambiense</i>	

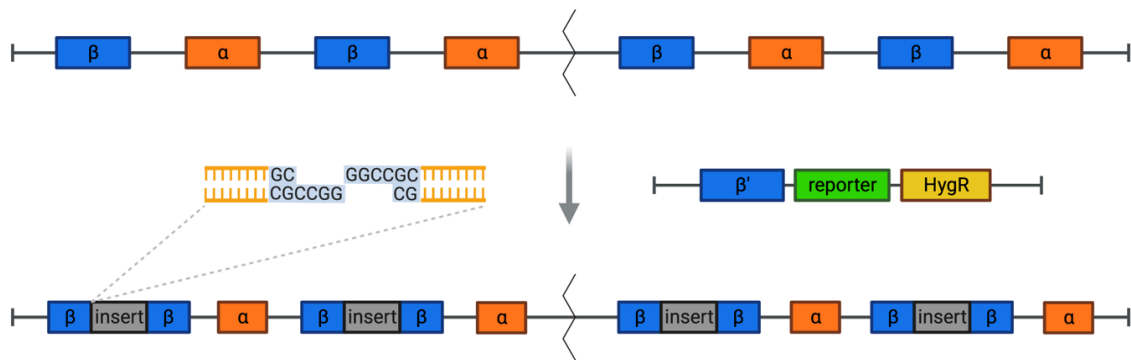
Supporting figure 1: Classification of trypanosomes [12].



Supporting figure 2: pHD 309 plasmid.

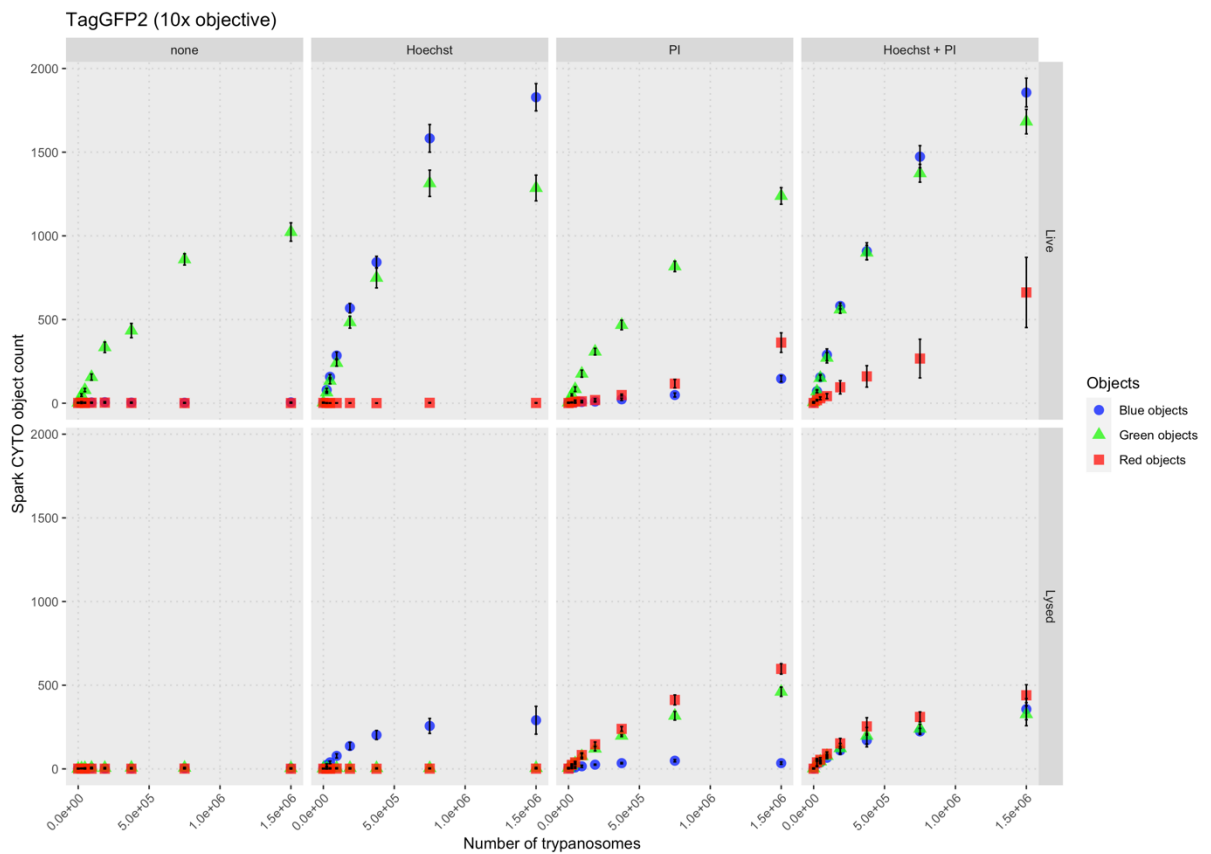


Supporting figure 3: pHD 309 plasmid with the integration of TagGFP2 and DsRed.



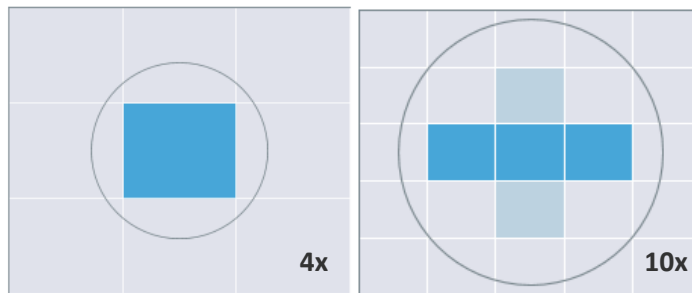
Supporting figure 4: Integration of reporter genes into the β -tubulin region.

Schematic representation of the integration TagGFP2 or DsRed and the hygromycin resistance gene into the β -tubulin repeats. The figure shows the α - and β -tubulin repeats on chromosome 1. The insert containing the β -tubulin region, fluorescent reporter gene and the hygromycin resistance gene (indicated as β' , reporter and HygR, respectively) is integrated via homologous recombination. Based on Jackson *et al.* [85], created with BioRender.com.



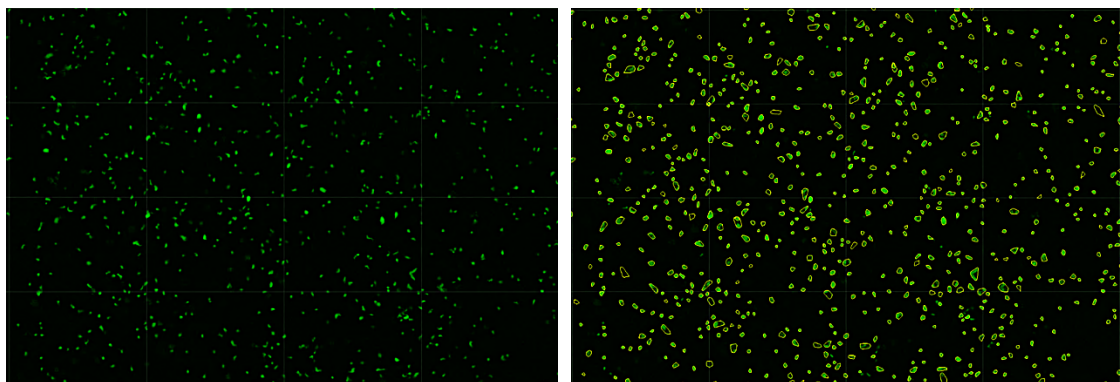
Supporting figure 5: Results for the 10x objective after adding Hoechst and PI to live and lysed AnTaR 1 cells.

The graph shows the objects counted by the Spark for the following conditions: no stain added, addition of Hoechst, PI, or a combination of both stains. The results were obtained simultaneously as the results shown in Figure 11. However, these counts were obtained using the 10x objective when focussing on the central part of the well. Overall, both graphs show the same pattern. Error bars indicate SD.



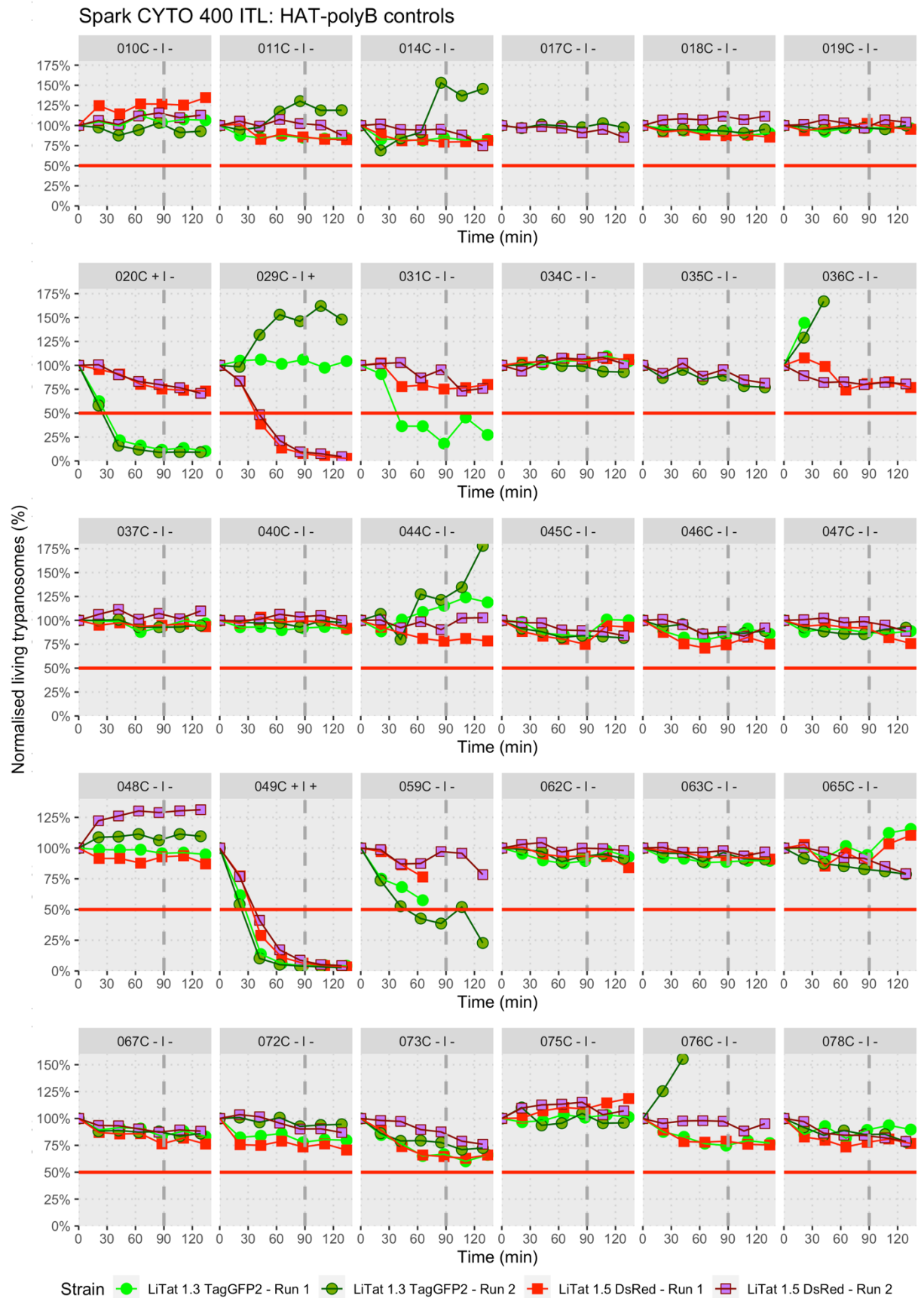
Supporting figure 6: Well pattern of the Spark CYTO 400.

Comparison of the selected imaging areas for the Spark CYTO 400 fluorescence imaging analysis. The blue area indicates the region chosen for imaging with the 4x objective, covering a larger area of the well compared to the three smaller regions (also highlighted in blue) selected for imaging with the 10x objective.



Supporting figure 7: Detection of the trypanosomes with the Spark CYTO 400 software.

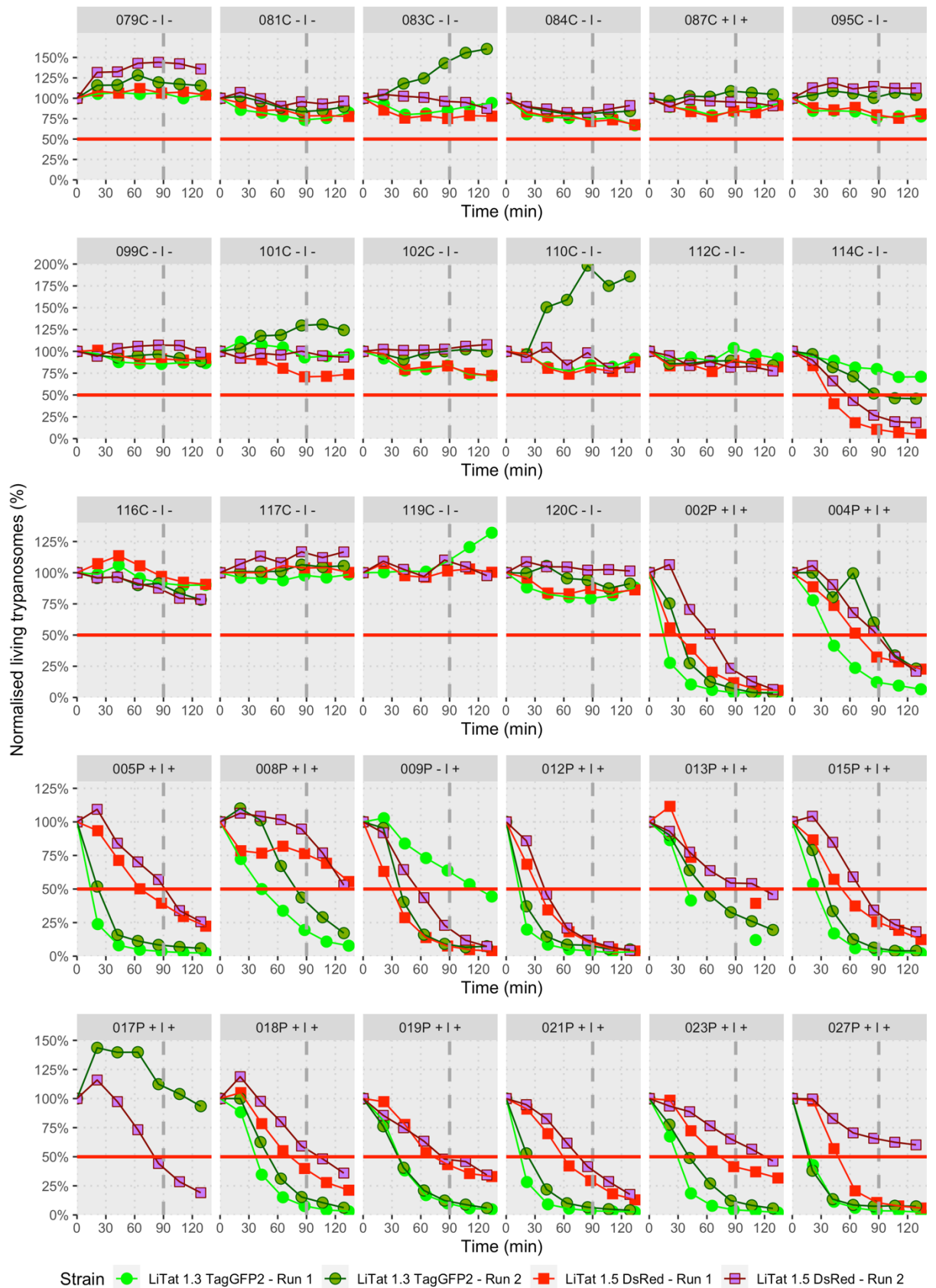
Illustration of the trypanosome counting process using the Spark CYTO 400. The first image displays the original image captured by the microscope, while the second image demonstrates the automated counting performed by the Spark's software. Yellow circles are generated by the software, surrounding the green objects that meet the specified criteria for trypanosome identification and counting.



Supporting figure 8: ITL results of HAT-PolyB samples (1).

Results of the kinetic ITL experiment. The symbols behind the sample ID indicate the original result of the ITL: “- I -” means negative for LiTat 1.3 and 1.5, “+ I -” means positive for LiTat 1.3 and negative for 1.5, etc. Trypanosomes counts were normalised based on their initial measurement. The dashed line shows the endpoint of the normal ITL (i.e., 90 min), while the red line shows the threshold of 50% lysis.

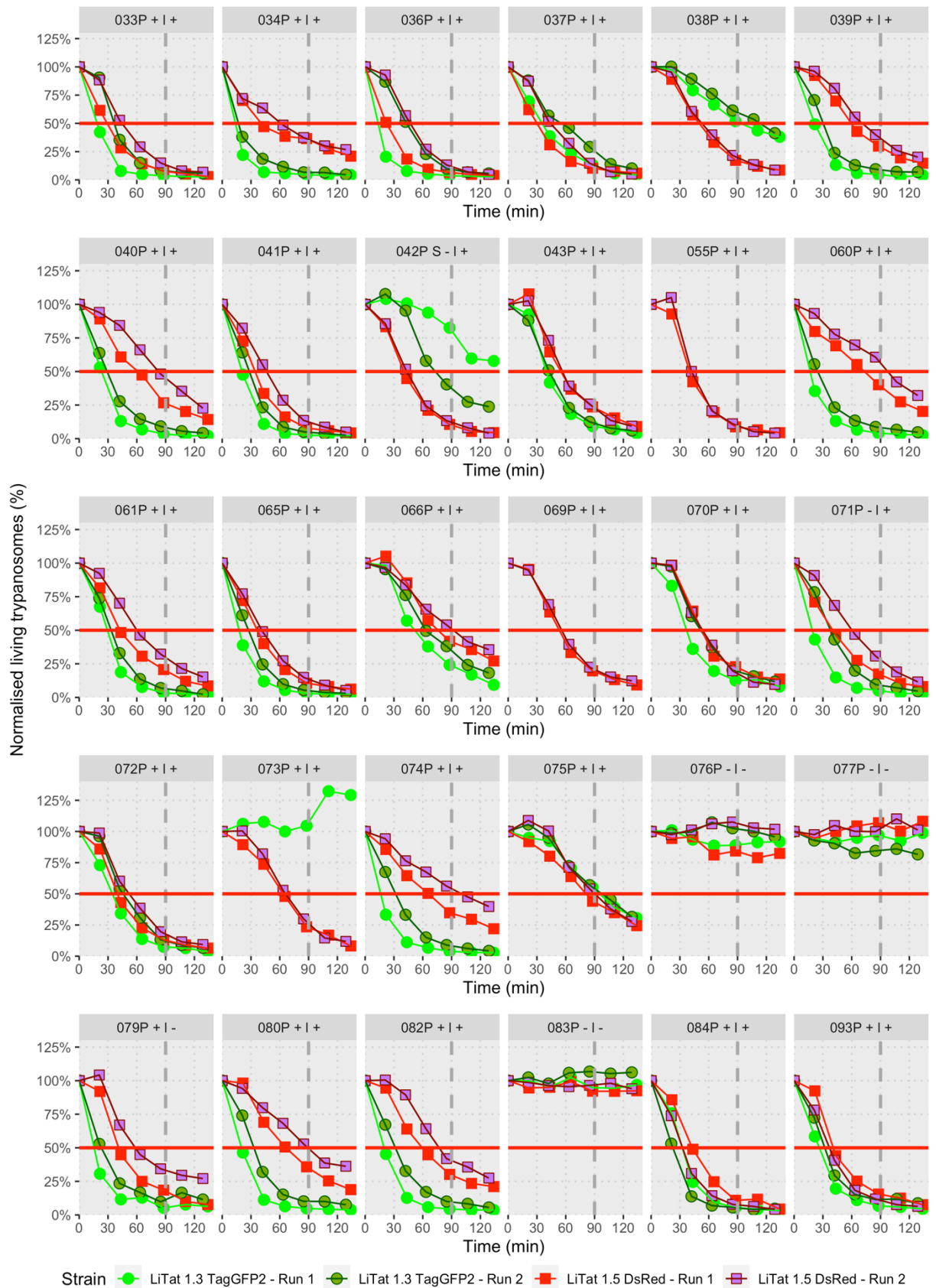
Spark CYTO 400 ITL: HAT-poly B controls and patients



Supporting figure 9: ITL results of HAT-PolyB samples (2).

Results of the kinetic ITL experiment. The symbols behind the sample ID indicate the original result of the ITL: “- I -” means negative for LiTat 1.3 and 1.5, “+ I -” means positive for LiTat 1.3 and negative for 1.5, etc. Trypanosomes counts were normalised based on their initial measurement. The dashed line shows the endpoint of the normal ITL (i.e., 90 min), while the red line shows the threshold of 50% lysis.

Spark CYTO 400 ITL: HATpoly-B patients



Supporting figure 10: ITL results of HAT-PolyB samples (3).

Results of the kinetic ITL experiment. The symbols behind the sample ID indicate the original result of the ITL: “- I -” means negative for LiTat 1.3 and 1.5, “+ I -” means positive for LiTat 1.3 and negative for 1.5, etc. Trypanosomes counts were normalised based on their initial measurement. The dashed line shows the endpoint of the normal ITL (i.e., 90 min), while the red line shows the threshold of 50% lysis.

Supporting table 1: The matching method in relation to the number of trypanosomes per volume [66].

Trypanosomes per field	Antilog (MM)	Trypanosomes per mL	Trypanosomes per well (50µL)
2	6.9	7.94×10^6	3.97×10^5
4	7.2	1.58×10^7	7.92×10^5
8	7.5	3.16×10^7	1.58×10^6
16	7.8	6.31×10^7	3.15×10^6
32	8.1	1.26×10^8	6.29×10^6
64	8.4	2.51×10^8	1.26×10^7
128	8.7	5.01×10^8	2.51×10^7
256	9.0	1.00×10^9	5.00×10^7